

Introduction

Problem Motivation

Air pollution forecasting is a major environmental and public health challenge. In this project, we design, implement, and evaluate machine learning models that predict air quality indicators using time-series sensor data. The primary aim is to model how pollutant concentrations evolve and forecast future pollution levels using historical observations and environmental factors.

Dataset Description

The dataset contains 9358 instances of hourly readings from metal oxide chemical sensors placed at road level. The readings were captured from a significantly polluted area in an Italian city and readings range from March 2004 to February 2005. The features consist of both sensor responses (PT08.S1–S5) and reference pollutant concentrations for carbon monoxide (CO), non-methane hydrocarbons (NMHC), benzene (C6H6), nitrogen oxides (NOx), and nitrogen dioxide (NO2). In addition, the dataset includes meteorological attributes such as temperature (T), relative humidity (RH), and absolute humidity (AH), which capture environmental factors influencing pollutant behaviour. Several variables contain missing values represented by the sentinel value -200 and were handled appropriately in the data cleaning process.

Project Goals

We break the problem down into a regression and classification problem, and we use a simple model such as linear/logistic regression, as well as more complex models like random forest regressors/classifiers. The main objectives for this report are Data analysis, Data pre-processing, feature engineering, outlier detection and residual analysis, model development and evaluation.

Data Analysis

Data Cleaning

The dataset contained substantial missing data, especially in NMHC(GT), where around 90% of values were absent. Other columns had roughly 16% missing entries as seen below.

	count	mean	std	min	25%	50%	75%	max
CO(GT)	7674.0	2.1527	1.4533	0.1000	1.1000	1.8000	2.9000	11.900
PT08.S1(CO)	8991.0	1099.8332	217.0800	647.0000	937.0000	1063.0000	1231.0000	2040.000
NMHC(GT)	914.0	218.8118	204.4599	7.0000	67.0000	150.0000	297.0000	1189.000
C6H6(GT)	8991.0	10.0831	7.4498	0.1000	4.4000	8.2000	14.0000	63.700
PT08.S2(NMHC)	8991.0	939.1534	266.8314	383.0000	734.5000	909.0000	1116.0000	2214.000
NOx(GT)	7718.0	246.8967	212.9792	2.0000	98.0000	180.0000	326.0000	1479.000
PT08.S3(NOx)	8991.0	835.4936	256.8173	322.0000	658.0000	806.0000	969.5000	2683.000
NO2(GT)	7715.0	113.0913	48.3701	2.0000	78.0000	109.0000	142.0000	340.000
PT08.S4(NO2)	8991.0	1456.2646	346.2068	551.0000	1227.0000	1463.0000	1674.0000	2775.000
PT08.S5(O3)	8991.0	1022.9061	398.4843	221.0000	731.5000	963.0000	1273.5000	2523.000
T	8991.0	18.3178	8.8321	-1.9000	11.8000	17.8000	24.4000	44.600
RH	8991.0	49.2342	17.3169	9.2000	35.8000	49.6000	62.5000	88.700
AH	8991.0	1.0255	0.4038	0.1847	0.7368	0.9954	1.3137	2.231

(a) Before Pre-Processing

	count	mean	std	min	25%	50%	75%	max
CO(GT)	9357.0	2.1527	1.3161	0.1000	1.2000	2.1527	2.6000	11.900
PT08.S1(CO)	9357.0	1099.8332	212.7917	647.0000	941.0000	1075.0000	1221.0000	2040.000
NMHC(GT)	914.0	218.8118	204.4599	7.0000	67.0000	150.0000	297.0000	1189.000
C6H6(GT)	9357.0	10.0831	7.3027	0.1000	4.6000	8.6000	13.6000	63.700
PT08.S2(NMHC)	9357.0	939.1534	261.5602	383.0000	743.0000	923.0000	1105.0000	2214.000
NOx(GT)	9357.0	246.8967	193.4266	2.0000	112.0000	229.0000	284.0000	1479.000
PT08.S3(NOx)	9357.0	835.4936	251.7440	322.0000	666.0000	818.0000	960.0000	2683.000
NO2(GT)	9357.0	113.0913	43.9210	2.0000	86.0000	113.0913	133.0000	340.000
PT08.S4(NO2)	9357.0	1456.2646	339.3676	551.0000	1242.0000	1456.2646	1662.0000	2775.000
PT08.S5(O3)	9357.0	1022.9061	390.6123	221.0000	742.0000	983.0000	1255.0000	2523.000
T	9357.0	18.3178	8.6576	-1.9000	12.0000	18.3000	24.1000	44.600
RH	9357.0	49.2342	16.9748	9.2000	36.6000	49.2342	61.9000	88.700
AH	9357.0	1.0255	0.3958	0.1847	0.7461	1.0154	1.2962	2.231

(b) After Mean Imputation

Figure 1: Summary statistics before and after preprocessing

We evaluated several imputation strategies. Because removing rows would cause excessive data loss (particularly for NMHC). Forward-backward fill averaging was initially considered to preserve temporal trends, but long consecutive NaN segments made this approach infeasible. We instead used column means to fill missing values, knowing that this reduces variance and introduces bias by ignoring time series structure. Due to the extreme missingness in NMHC, rows after 2004-05 were excluded entirely to avoid imputing almost the entire column; its remaining 914 rows were then used for model training and testing.

Summary Statistics

The tables in figure 1 above show the change of the summary statistics of the variables before and after imputation respectively. We see that the mean of all variables is the same, which was expected since the filled values are heavily centralised with the existing mean. The median for all columns has increased since we increased the number of points $\geq$ mean. This also leads to the lower quartiles to increase and upper quartiles to decrease as they are now closer to the mean and median. The standard deviation has decreased after imputation, as more values are now closer to the mean.

Exploration of Patterns

Initially, we made a correlation heatmap to see the predictive power of the features and target variables and to check the multi-collinearity between features; this is attached in the appendix. We also made time series plots for all feature and target variables (appendix). One notable issue appears in the time series of `NMHC(GT)`, where we see a flat line after 2004-05 due to a long sequence of imputed values.

From the time series plots, we observe that pollutant concentrations tend to peak twice each year, around months 03–05 and 09–11, indicating a seasonal trend. We also generated scatter plots of target variables against the features, normalising values so that they are on the same scale. Most features show roughly linear relationships with the targets. However, `T`, `RH`, and `AH` show no clear relationship and form clusters. The scatter plots of one of the target variables `C6H6(GT)` is shown below.

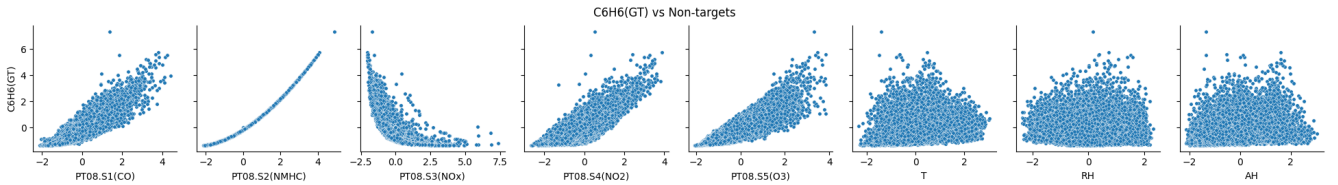


Figure 2: Scatter plot of `C6H6(GT)`

Distribution plots show that none of the variables follow a normal distribution, which is expected because pollutant concentrations are non-negative and often right-skewed. The skewness observed in several variables guided our decision to apply transformations where appropriate. We hypothesised that because the distributions are not normal, the residuals from the linear models will not follow a normal distribution

Anomaly Detection

We used an unsupervised isolation forest to detect anomalies, tuning its contamination parameter based on pseudo-ground truth labels created with a modified Z-score method. We focused on maximizing precision to ensure that flagged anomalies truly correspond to pollution events, rather than chasing higher recall, which would increase false alarms.

Looking at how different features relate to anomalies, we found that weekends had little effect, while temperature showed a clear impact: `CO` and `C6H6` tended to increase with higher temperatures, `NMHC` and `NO2` decreased, and `NOx` spiked during temperature extremes. We made the data more robust by removing these anomalies, specifically for `CO`, where the cleaned dataset led to lower MAE and better generalisation. However, this approach isn't perfect: in some cases, removing anomalies might also eliminate meaningful rare events that the model needs to recognise.

Methodology

Preprocessing

We merged date and time into a single variable called timestamp, and we added new features such as hour, weekday and month. We normalised continuous features for time series plotting, since all series will be on the same scale of units.

Feature Engineering

To capture temporal structure in the data, we appended lag features of the target variables at multiple horizons (e.g. 1, 3, 6, 12, and 24 hours). We also added moving averages over short and longer windows (3, 6, 12, 24 hours). The `PT08.S3(NOx)` variable was seen to be have a strong inverse relationship with the target variables, and `PT08.S2(NNMHC)` variable seemed to have a slight curved relationship with the target variables. Thus, we transformed these using $1/(x^2 + \epsilon)$, and x^2 respectively. Note that the ϵ is a small positive constant was added to prevent introducing inf values.

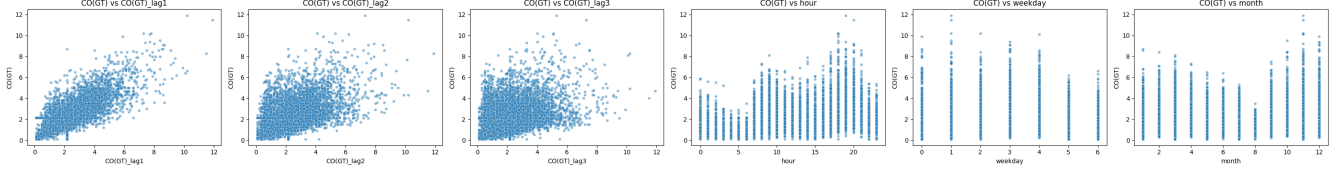


Figure 3: CO(GT) vs new time derived features

The derived features hour, weekday and month had cyclical patterns with the target variables, as seen in the figure above. Thus, to make these suitable for a linear model, we derived binned features to linearly approximate the cyclical behaviour. There were 4 month bins, each being 3 months long, which captured the seasonal features. There were 4 hour bins, each being 6 hours long. This captured the daily dynamics of public transportation during rush hours. We considered the same for weekday feature, however there was no clear pattern that held accross all targets, and so we created only 1 derived bin called `is_sunday`. This was because at weekends, the concentration was lowest accross all pollutants. This was expected since majority of the population relaxes at home during weekends, and thus there is less emission from vehicles.

Model Selection Rationale

We considered linear random forest models for the regression and classification problems, with the rational of comparing performance with increasing model complexity. We did not considered more complex models such as neural networks, because we wanted a balance between accuracy and interprtability. The most basic and rather trivial model was the naive regressor/classifier which provides a baseline which our more complex models can compare against. We used ridge regularisation for the linear models, and observed the coefficients of the features to see which predictors were most relevant. We also used feature importance for random forests to extract which features carry the most weight for the model's predictions.

Results

Table 1: Baseline RMSE

Target	t+1	t+6	t+12	t+24
CO(GT)	0.824	1.769	1.767	1.283
NMHC(GT)	156.138	291.945	256.317	179.857
C6H6(GT)	3.613	8.133	8.378	5.887
NOx(GT)	114.070	270.160	265.145	199.852
NO2(GT)	27.182	67.571	67.871	43.953

Table 2: Baseline accuracy

Horizon	Naive test accuracy
t + 1	0.739
t + 6	0.398
t + 12	0.377
t + 24	0.577

horizon	train_acc	cv_acc	test_acc	Precision	Recall	F1
t+1	0.925447	0.791047	0.739982	0.727185	0.728753	0.727933
t+6	0.888873	0.671791	0.607764	0.589913	0.599747	0.592581
t+12	0.880855	0.626351	0.591499	0.573155	0.584674	0.574366
t+24	0.674638	0.608953	0.544759	0.584747	0.556845	0.551880

Figure 4: Random forest classifier

	horizon	train_acc	cv_acc	test_acc	Precision	Recall	F1
0	t+1	0.757209	0.717399	0.689671	0.703861	0.694992	0.692763
1	t+6	0.611760	0.539527	0.464525	0.537188	0.486292	0.473061
2	t+12	0.587284	0.525000	0.446532	0.564204	0.475097	0.443559
3	t+24	0.621466	0.583784	0.486730	0.568183	0.512917	0.494470

Figure 5: Logistic Regression Classifier

The residual plots and other useful plots are attached in the appendix.

Table 3: Linear Regression RMSE

Target	t+1	t+6	t+12	t+24
CO(GT)	0.721	1.272	1.260	1.234
NMHC(GT)	124.03	168.937	153.240	164.069
C6H6(GT)	3.261	5.662	6.039	5.563
NOx(GT)	107.803	192.393	187.548	182.187
NO2(GT)	26.279	44.033	42.935	39.810

Table 4: Random Forest Regression RMSE

Target	t+1	t+6	t+12	t+24
CO(GT)	0.757	1.267	1.329	1.260
NMHC(GT)	120.331	180.723	172.517	165.839
C6H6(GT)	3.403	5.982	6.340	5.651
NOx(GT)	100.829	195.516	213.608	187.443
NO2(GT)	28.717	46.011	46.144	39.950

Discussion

Interpretation of Results

Linear Regression

We see that the linear regression model improves significantly upon the naive baseline model across all target variables. We observe that the RMSE per horizon follows a similar trend to the baseline's. More specifically, we see that the RMSE is lowest at horizon $t + 1$, and highest at time $t + 6$. We also observe that the highest variance of RMSE occurs for the NOx(GT) target variable. One of the possible reasons for this could be that NOx(GT) just has higher values in general. However, when we consult the time series plot of NOx(GT) in the appendix, we observe that the deviations for the concentration was relatively low in 2004, and it increases significantly in the year 2005. We can derive the feature importance of each variable from the raw coefficients. For the CO(GT) target at $t + 1$ horizon, we observe that the concentration at time t has the most importance. This was also observed previously with the scatter plot, where we saw a strong correlation between CO(GT) and lag 1.

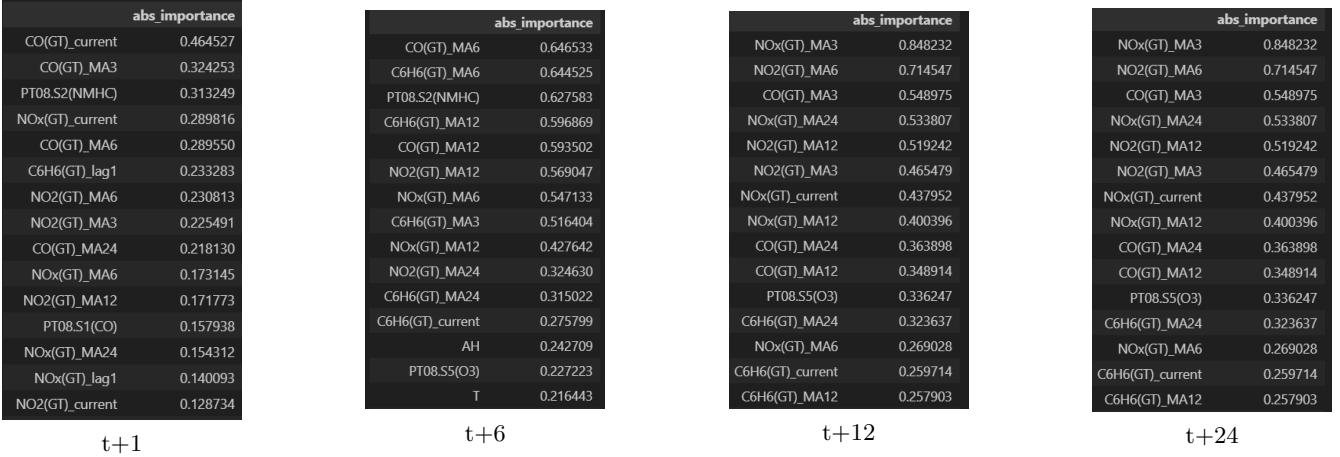


Figure 6: Feature importance for the linear regression CO(GT) at different prediction horizons.

We revisit the distribution plots, and we observe that none of the target variables have a normal or lognormal distribution except for NO2. This is a violation of the linear model assumptions, as we assume that the target follows a normal distribution (or $\log(\text{target})$ follows normal distribution). We have also attached the residual plots for linear regression in the appendix. We see that for the residuals ϵ , $\mathbb{E}[\epsilon] = 0$ across all targets. However, we do not see $\text{Var}[\epsilon] = \sigma^2$, that is to say the variance of the residuals are not constant. This violates the assumptions of linear regression.

Logistic Regression

We see a significant drop in accuracy of 0.69 at $t + 1$ to 0.46 at $t + 6$. This is unexpectedly a worse performance than the baseline. The recall and precision across classes and time horizons is shown in the figure below. We see that the precision drops across horizons. We see that mid class consistently has a high FP, suggesting that the model is biased against this class. The high class is consistently predicted well, with fewer FP relative to TP. We visit the coefficients table in the appendix, and we observe that the sensor variables have the higher coefficients across most classes and horizons. The coefficient of CO(GT) at time t is about 0.75 for class high and horizon $t + 1$. This means that an increase in 1 unit of concentration of CO(GT) at time t , increases the log odds of it being a high class concentration at time $t + 1$ by $\exp(0.75)$.

The correlation heatmap attached in the appendix, suggests multicollinearity, since a lot of the predictors, specifically the sensor variables are strongly correlated. This violates the assumptions of logistic classifier and linear regression. We can improve upon this by using principal component analysis to reduce dimensionality and produce uncorrelated predictors. We could also omit the weakly correlated predictors during feature selection. The linear models (both regression and classification) also inherently assume linear relationship between predictors and targets. However, because this is a time series data set, the relationships between predictors and targets constantly evolve with time and a linear model cannot sufficiently capture that.

	horizon	class	TP	FP	FN	TN
0	1	low	642	128	324	1152
1	1	mid	424	437	232	1153
2	1	high	483	132	141	1490
3	6	low	319	145	646	1131
4	6	mid	415	846	238	742
5	6	high	307	209	316	1409
6	12	low	250	76	710	1199
7	12	mid	501	989	151	594
8	12	high	247	172	376	1440
9	24	low	317	108	643	1155
10	24	mid	439	826	204	754
11	24	high	326	207	294	1396

Figure 7: Confusion matrix statistics

Random Forest Regressor

The residual plots show that for most horizons, errors are generally centered around zero, indicating no substantial bias in predictions. Larger deviations are observed during periods of high pollutant concentration, suggesting the model struggles with extreme events. This is reinforced by the model’s inability to match observed spikes in the time series plots. The model performs noticeably better than the naive baseline for the $t+6$ and $t+12$ horizons, but only marginally better for $t+1$ and $t+24$. This pattern can be attributed to the naive baseline’s effectiveness at short horizons ($t+1$) due to reliance on the most recent observation, and at $t+24$ due to the cyclic nature of temperature and pollutant patterns.

Random Forest Classifier

In order to find suitable hyperparameters, random search was used with 5-fold cross validation. Initially, the model was severely overfitting the data, however after adjusting the hyperparameter space that random search uses, this limited the amount of overfitting encountered, however, it still remains apparent. Some examples of the approaches used to mitigate overfitting include reducing the number of trees, reducing the max depth of each tree, reducing the number of features in each tree, and increasing the size of leaf nodes. In comparison to the naive baseline, the random forest classifier appears to show better accuracy for $t+6$ and $t+12$, but a comparable accuracy for $t+1$ and $t+24$. When looking at the feature importances, $t+1$ and $t+24$ show that CO(GT) has the highest importance whereas this is not the case for $t+6$ and $t+12$ which may be an indication of the comparable accuracies for $t+1$ and $t+24$. The confusion matrix attached in the appendix. For each time horizon, we see that the model is best at classifying the low label. A further look reveals that the model’s majority classifications are also in the low class, and so it favours that class. We also see that as the time horizon gets larger, the model classifies more instances into mid class, i.e. the model favours the mid class more and more.

feature	importance	feature	importance	feature	importance	feature	importance	
0	CO(GT)	0.250526	13	hour	0.243324	13	hour	0.198831
7	NO2(GT)	0.107427	19	CO(GT)_ma_24	0.114806	18	CO(GT)_ma_12	0.121121
5	NOx(GT)	0.094981	0	CO(GT)	0.109411	19	CO(GT)_ma_24	0.101118
4	PT08.S2(NMHC)	0.082845	17	CO(GT)_ma_6	0.070522	16	CO(GT)_ma_3	0.093168
20	CO(GT)_lag1	0.072157	16	CO(GT)_ma_3	0.065072	14	weekday	0.074467
3	C6H6(GT)	0.069170	14	weekday	0.061784	17	CO(GT)_ma_6	0.051819
16	CO(GT)_ma_3	0.046349	5	NOx(GT)	0.048195	10	T	0.043759
13	hour	0.035427	20	CO(GT)_lag1	0.046591	11	RH	0.038567
1	PT08.S1(CO)	0.034419	18	CO(GT)_ma_12	0.045042	0	CO(GT)	0.037114
17	CO(GT)_ma_6	0.026678	23	NOx(GT)_lag1	0.023754	20	CO(GT)_lag1	0.035773
22	C6H6(GT)_lag1	0.022118	11	RH	0.022524	12	AH	0.024843
18	CO(GT)_ma_12	0.018418	10	T	0.020829	23	NOx(GT)_lag1	0.021036
19	CO(GT)_ma_24	0.017462	7	NO2(GT)	0.016784	5	NOx(GT)	0.020482
23	NOx(GT)_lag1	0.016227	12	AH	0.015229	9	PT08.S5(O3)	0.017576
24	NO2(GT)_lag1	0.015213	15	month	0.014217	8	PT08.S4(NO2)	0.017400
t+1			t+6			t+12		

feature	importance	feature	importance
0	CO(GT)	0.265251	
14	weekday	0.095082	
13	hour	0.073440	
4	PT08.S2(NMHC)	0.064505	
3	C6H6(GT)	0.052143	
20	CO(GT)_lag1	0.046275	
16	CO(GT)_ma_3	0.043938	
23	NOx(GT)_lag1	0.040945	
5	NOx(GT)	0.039266	
24	NO2(GT)_lag1	0.038414	
19	CO(GT)_ma_24	0.026701	
10	T	0.026532	
12	AH	0.023463	
18	CO(GT)_ma_12	0.022669	
1	PT08.S1(CO)	0.021577	
t+24			

Figure 8: Feature importance for the random forest classifier at different prediction horizons.

Limitations

The models assume access to GT values, which may not always be feasible in deployment. While it is possible to train using only sensor data, some sensors, such as NO₂ and NOx, exhibit low correlation with the GT values. Using GT features ensures methodological consistency with the naive baseline. Sensor drift over time may degrade accuracy, potentially requiring periodic retraining or replacement of sensors. Additionally, temporal concept drift is a concern since training was performed on one year of data, and trends may vary across years. This necessitates retraining the model on newly collected data to maintain performance.

Improvement Strategies

Feature engineering included lag variables, moving averages, and cyclic factors. Further improvements could involve constructing non-linear features, such as combining traffic activity with environmental measurements, to better capture complex interactions. Outliers were retained to expose the model to real-world conditions, but a weighted training strategy could reduce their impact. Although class imbalance is relatively minor, with low, medium, and high concentrations comprising 33.9–43%, 29–40.5%, and 25.6–27.8% respectively, addressing this imbalance could improve classification performance.

Conclusion

Summary of Findings

Pollutant concentrations display strong temporal patterns, with short-term dependencies being particularly influential. The linear regression model improves upon the naive baseline overall, but its performance declines as the prediction horizon extends. Moving averages, especially for CO(GT), consistently contribute the most predictive power. Violations of linear model assumptions, such as non-normal distributions and heteroscedasticity, limit the ability of linear models to capture the full complexity of the data. The random forest regressor, while theoretically capable of modeling non-linear trends, tends to overfit noisy data, resulting in performance comparable to the linear model. Classification models indicate that high pollutant events are more difficult to forecast, with both linear and tree-based models favoring low and mid concentrations over longer horizons. Logistic regression and random forests maintain better accuracy for t+6 and t+12 compared to the naive baseline, reflecting the strong daily periodicity of CO(GT) at certain time horizons.

Implications

These results underscore the importance of accounting for temporal dependencies, non-linear relationships, and interactions among features when forecasting air pollution levels. Short-term forecasts rely heavily on linear trends and lagged features, whereas capturing extreme events or long-term trends may require ensemble or non-linear models. Careful tuning and strategies to mitigate overfitting are essential to ensure robustness and reliable deployment of predictive models.

References

Exploring Machine Learning Approaches for Time Series. <https://www.geeksforgeeks.org/machine-learning/exploring-machine-learning-approaches-for-time-series/>

GeeksforGeeks (2019a). Python — Linear Regression using sklearn. [online] GeeksforGeeks. Available at: <https://www.geeksforgeeks.org/machine-learning/python-linear-regression-using-sklearn/>

Anomaly detection using Isolation Forest. [online] GeeksforGeeks. Available at: <https://www.geeksforgeeks.org/machine-learning/anomaly-detection-using-isolation-forest/>. Peter, M., Aldo, D., Cheng, F. and Ong, S. (2020).

MATHEMATICS FOR MACHINE LEARNING. [online] Available at: <https://mml-book.github.io/book/mml-book.pdf>.

Scikit Learn (n.d.). sklearn.ensemble.IsolationForest — scikit-learn 0.23.1 documentation. [online] scikit-learn.org. Available at: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.IsolationForest.html>. Scikit-Learn (2025).

sklearn.ensemble.RandomForestClassifier — scikit-learn 0.20.3 Documentation. [online] Scikit-learn.org. Available at: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>.

scikit-learn (2014). LogisticRegression. [online] Scikit-learn.org. Available at: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html.

scikit-learn (2018). 3.2.4.3.2. sklearn.ensemble.RandomForestRegressor — scikit-learn 0.20.3 documentation. [online] Scikit-learn.org. Available at: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html>.

scikit-learn developers (2019). sklearn.linear_model.LinearRegression | scikit-learn 0.22 documentation. [online] Scikit-learn.org. Available at : .

scikit-learn. (2024). Linear Regression Example. [online] Available at: https://scikit-learn.org/1.5/auto_examples/linear_model/plot_ols.html.

Willems, K. (2021). Matplotlib Cheat Sheet: Plotting in Python. [online] Datacamp.com. Available at: <https://www.datacamp.com/cheat-sheet/matplotlib-cheat-sheet-plotting-in-python>.

Appendix

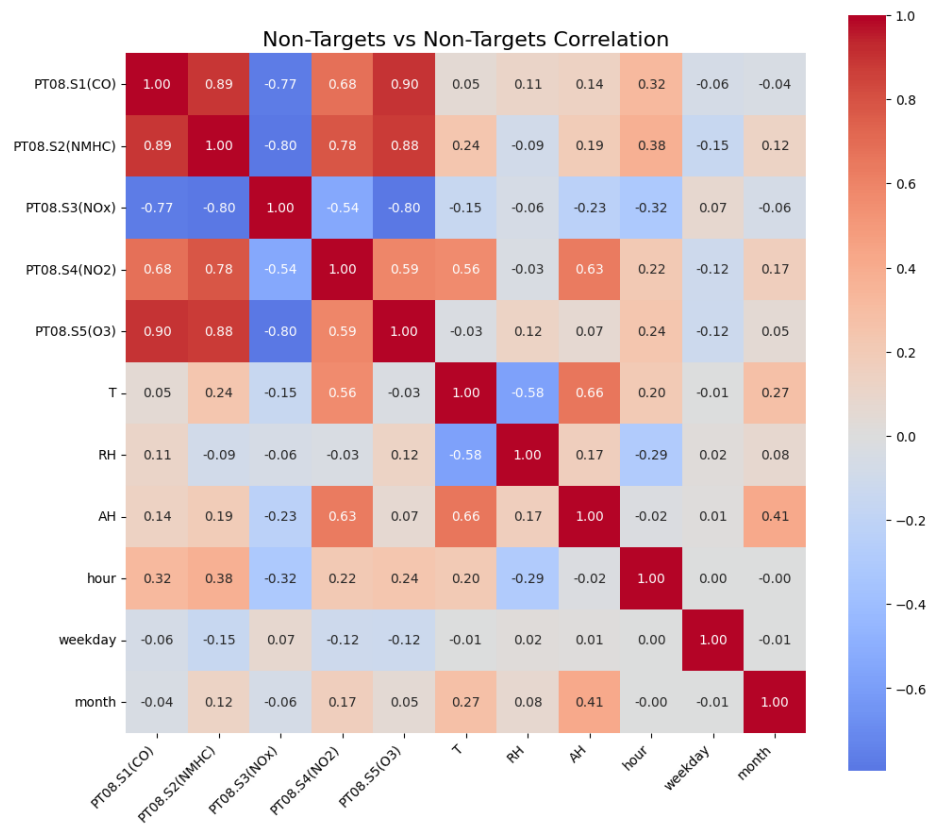


Figure 9: Feature-Feature correlation

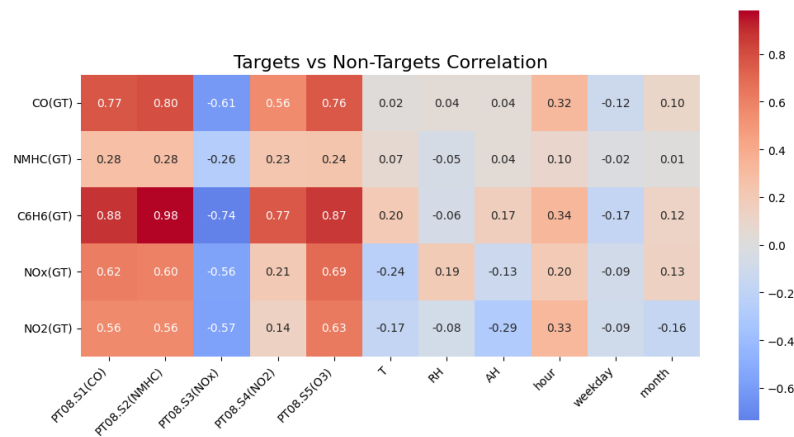


Figure 10: Target-Feature correlation

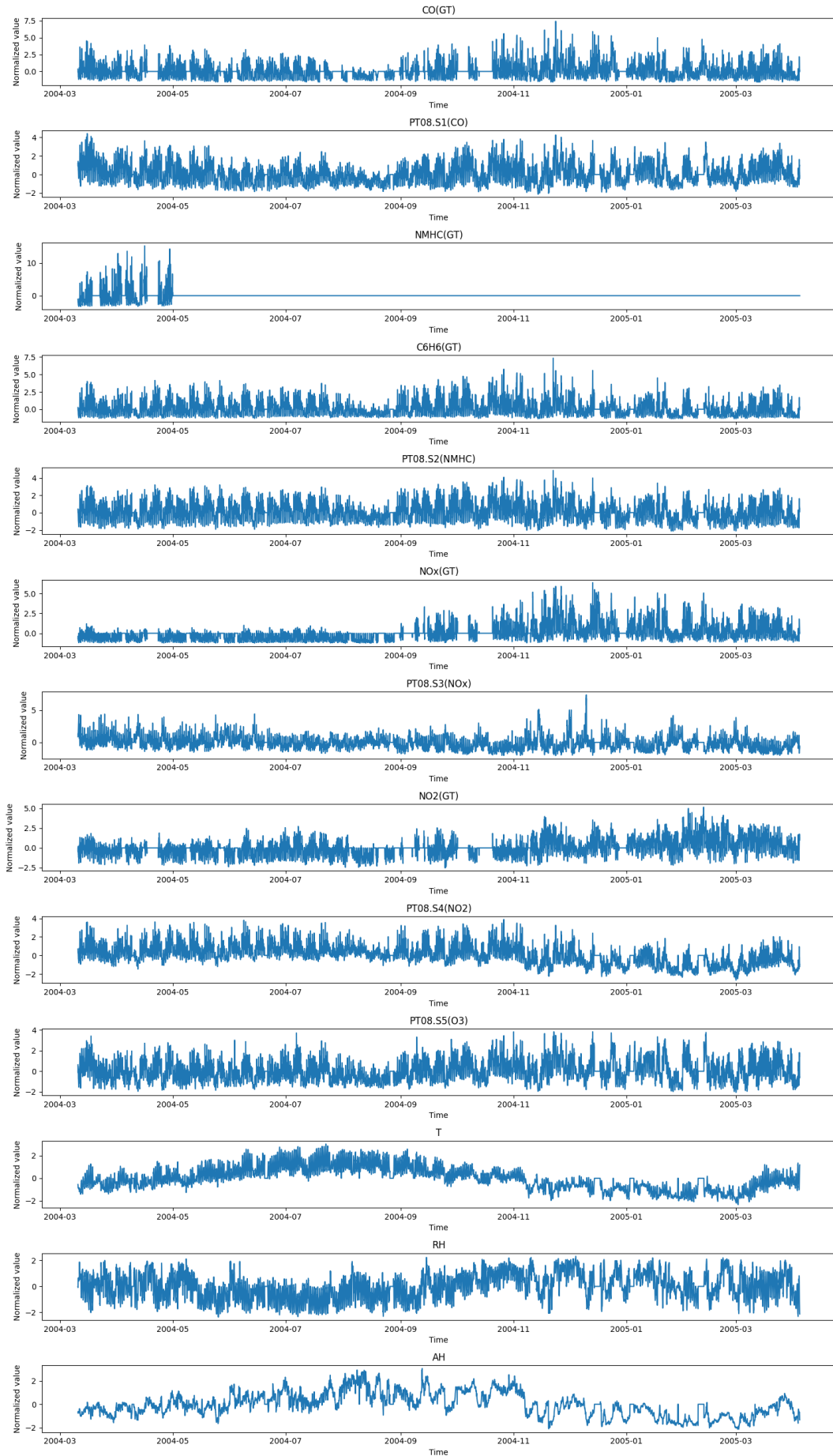


Figure 11: Normalised Time Series of each variable

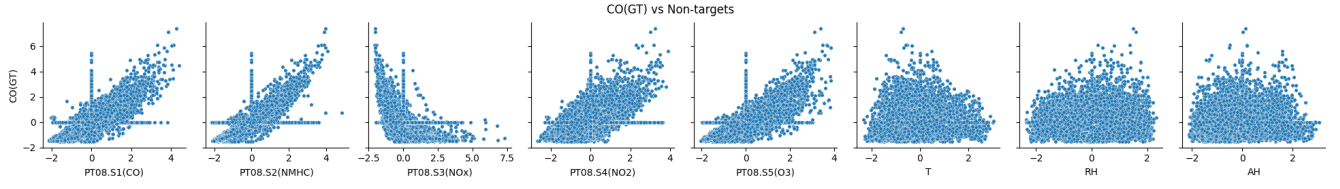


Figure 12: Scatter plot of CO(GT)

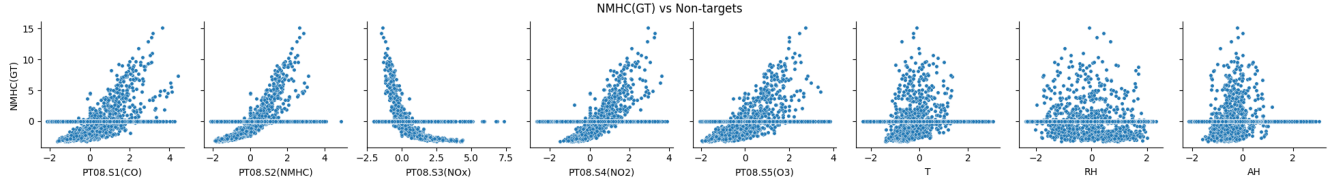


Figure 13: Scatter plot of NMHC(GT)

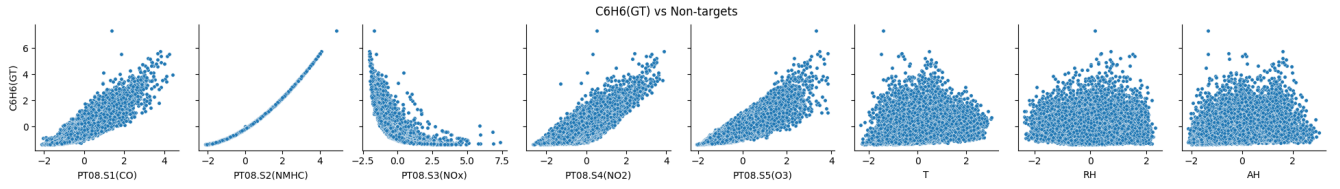


Figure 14: Scatter plot of C6H6(GT)

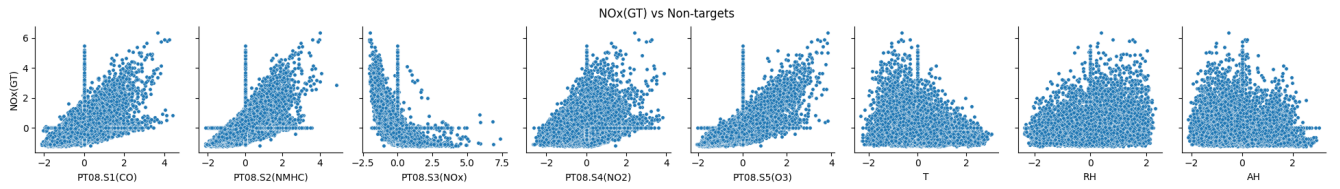


Figure 15: Scatter plot of NOx(GT)

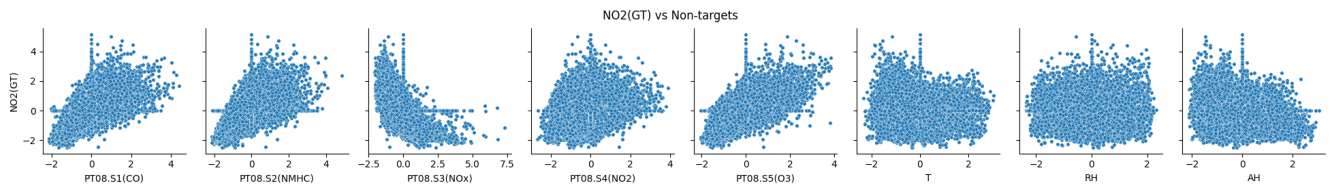


Figure 16: Scatter plot of NO2(GT)

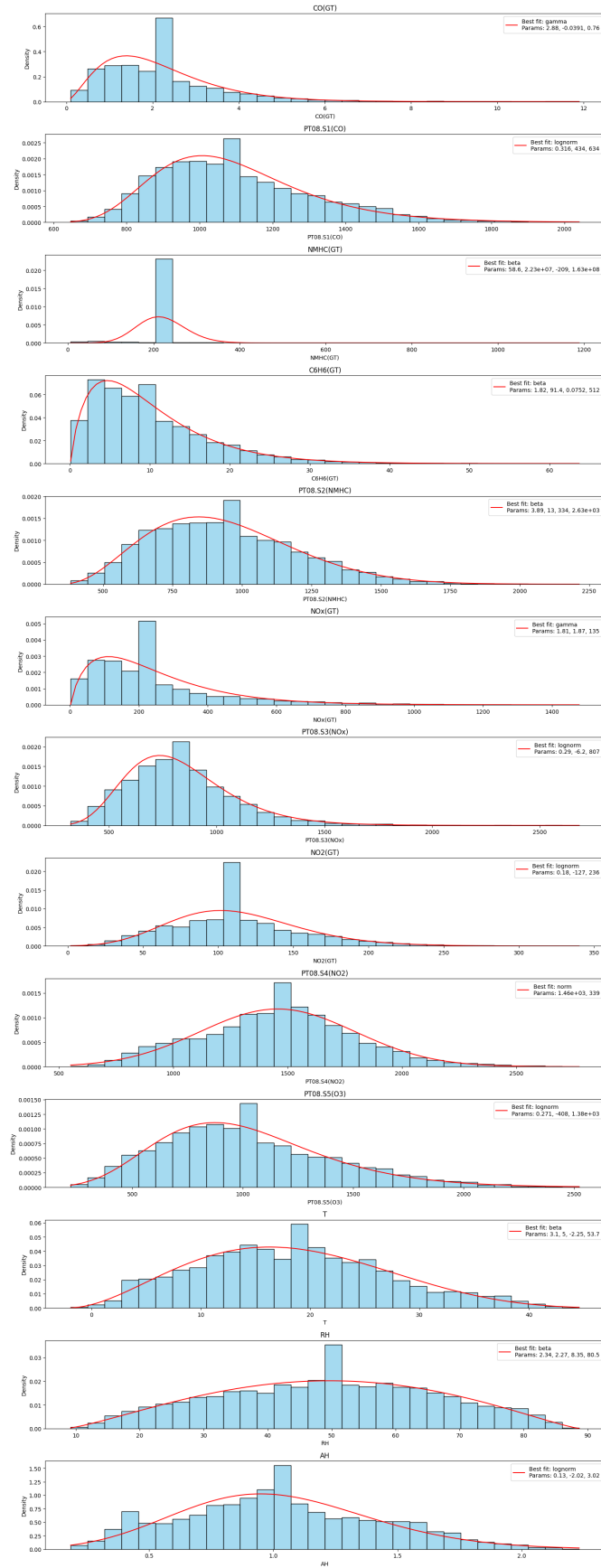


Figure 17: Distribution of variables

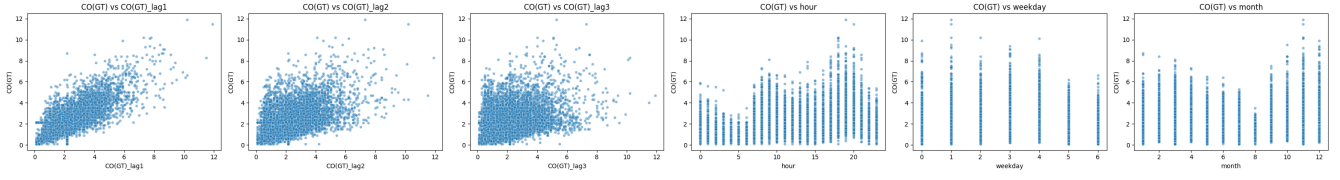


Figure 18: CO(GT) vs new time derived features

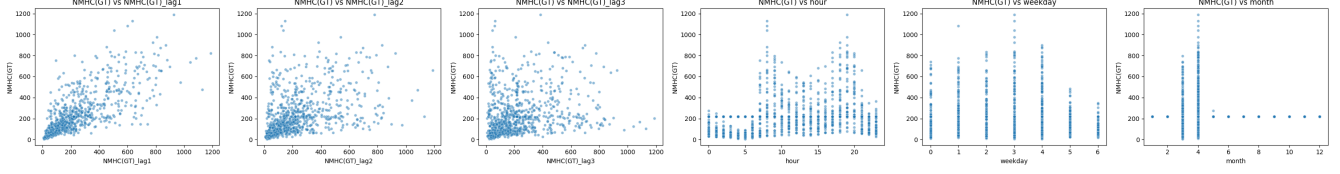


Figure 19: NMHC(GT) vs new time derived features

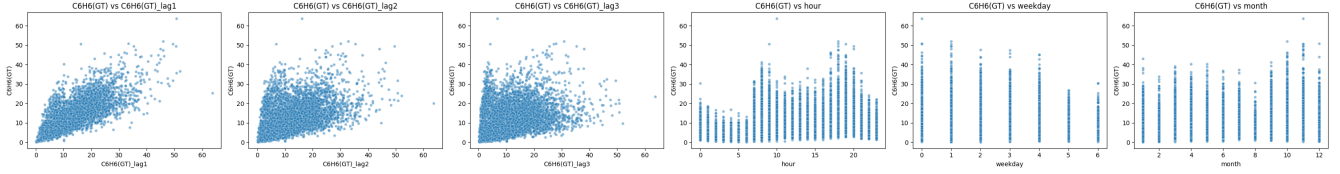


Figure 20: C6H6(GT) vs new time derived features

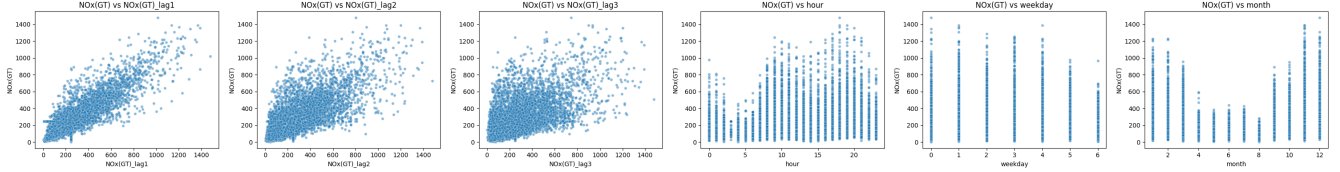


Figure 21: NOx(GT) vs new time derived features

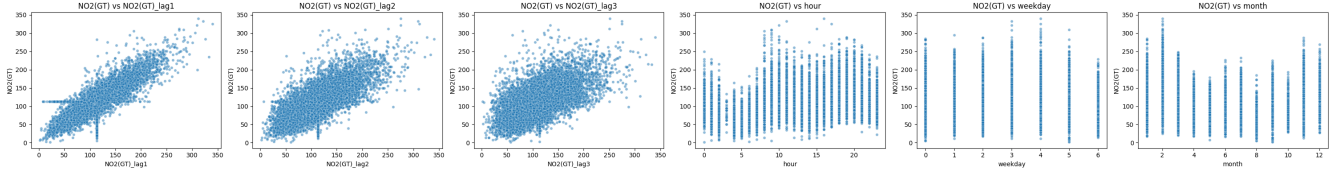


Figure 22: NO2(GT) vs new time derived features

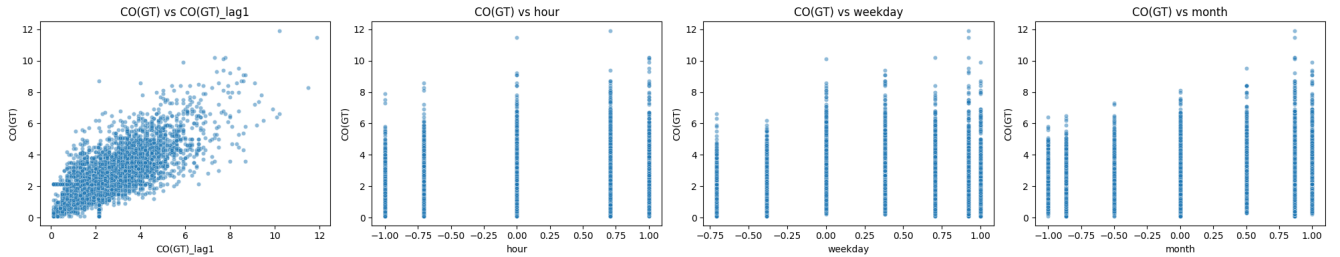


Figure 23: CO(GT) vs transformed hour,weekday,month

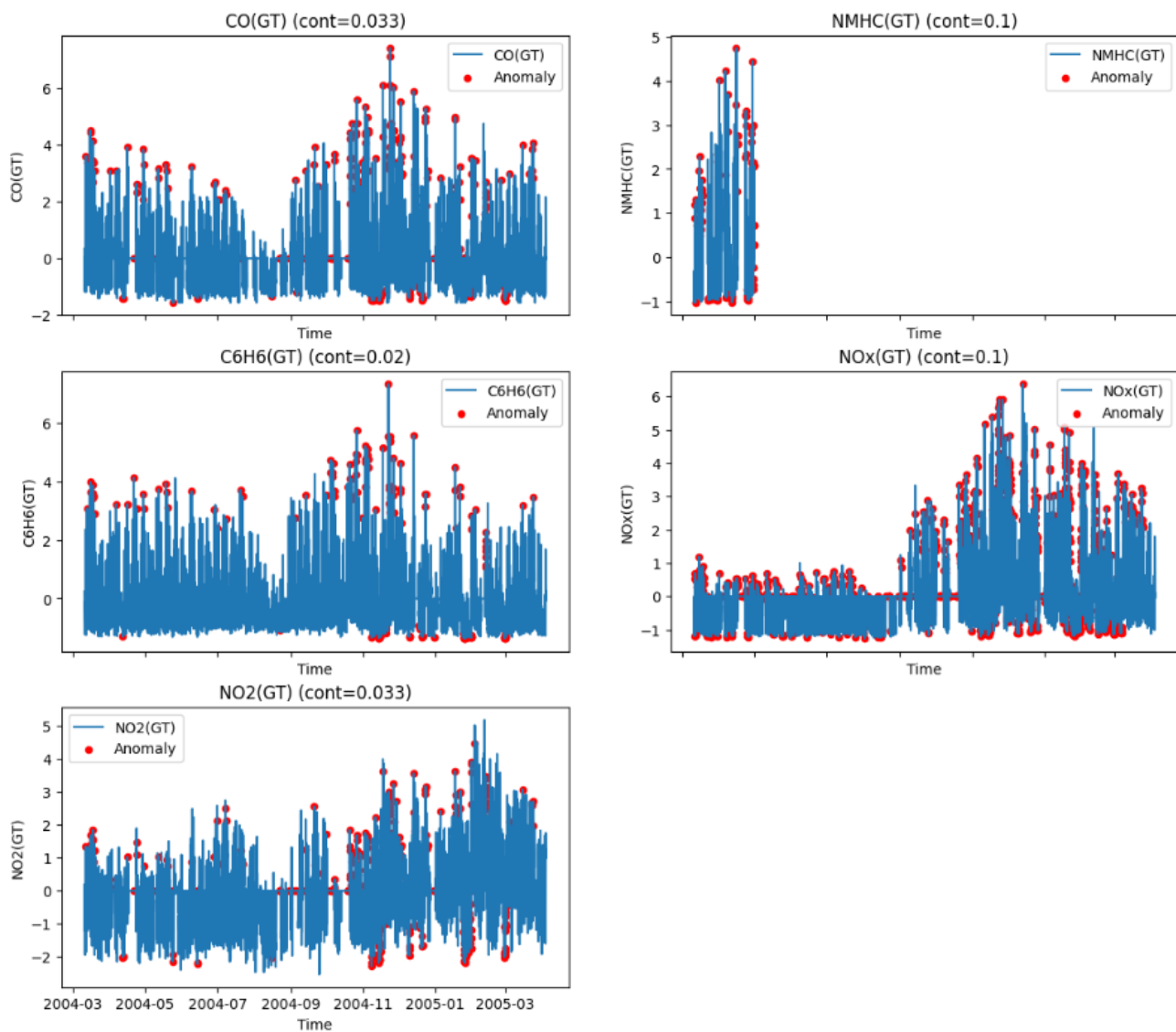


Figure 24: Time series graphs for all targets with anomalies flagged

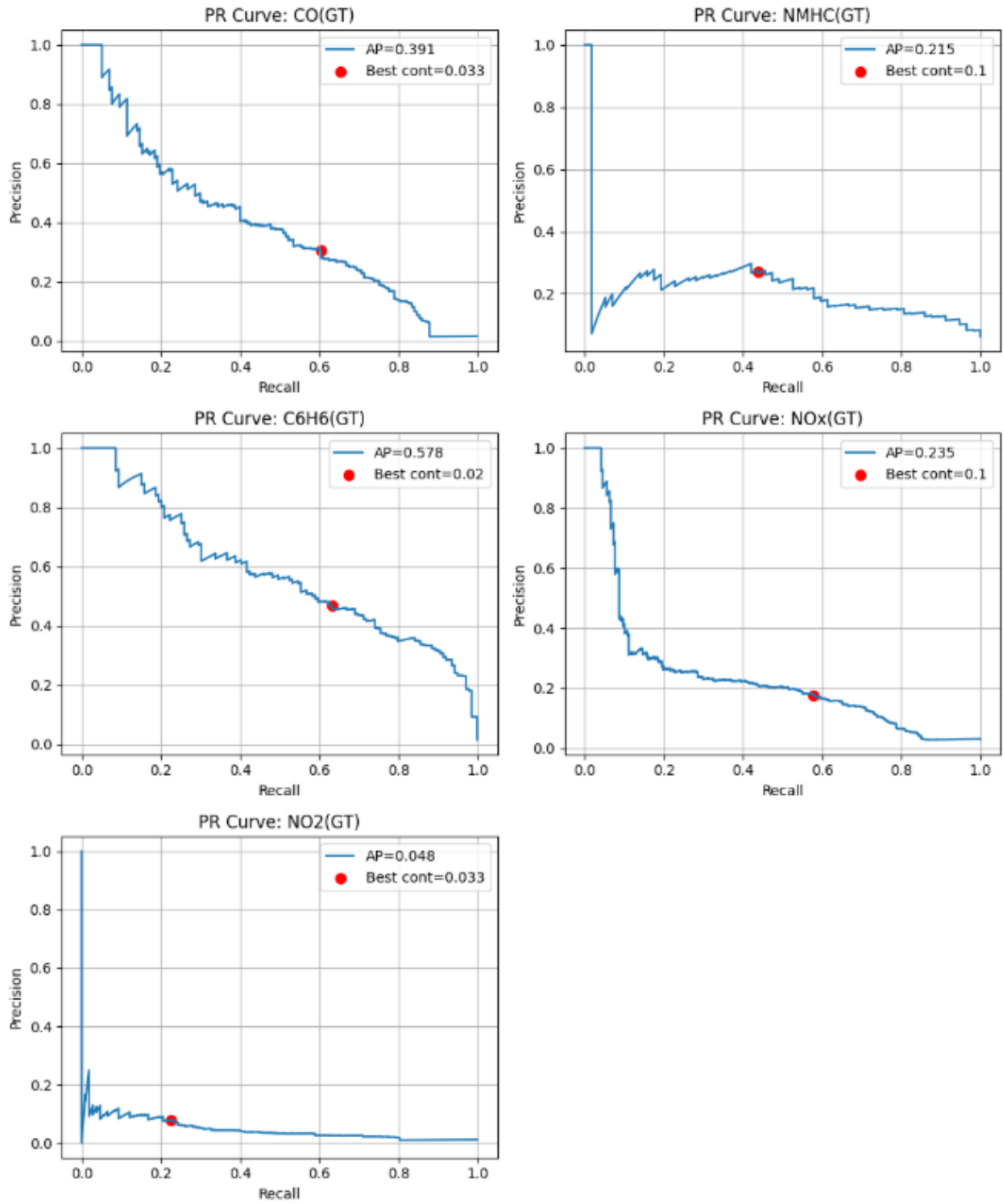


Figure 25: Precision-recall curves for all targets, with point of best contamination labelled

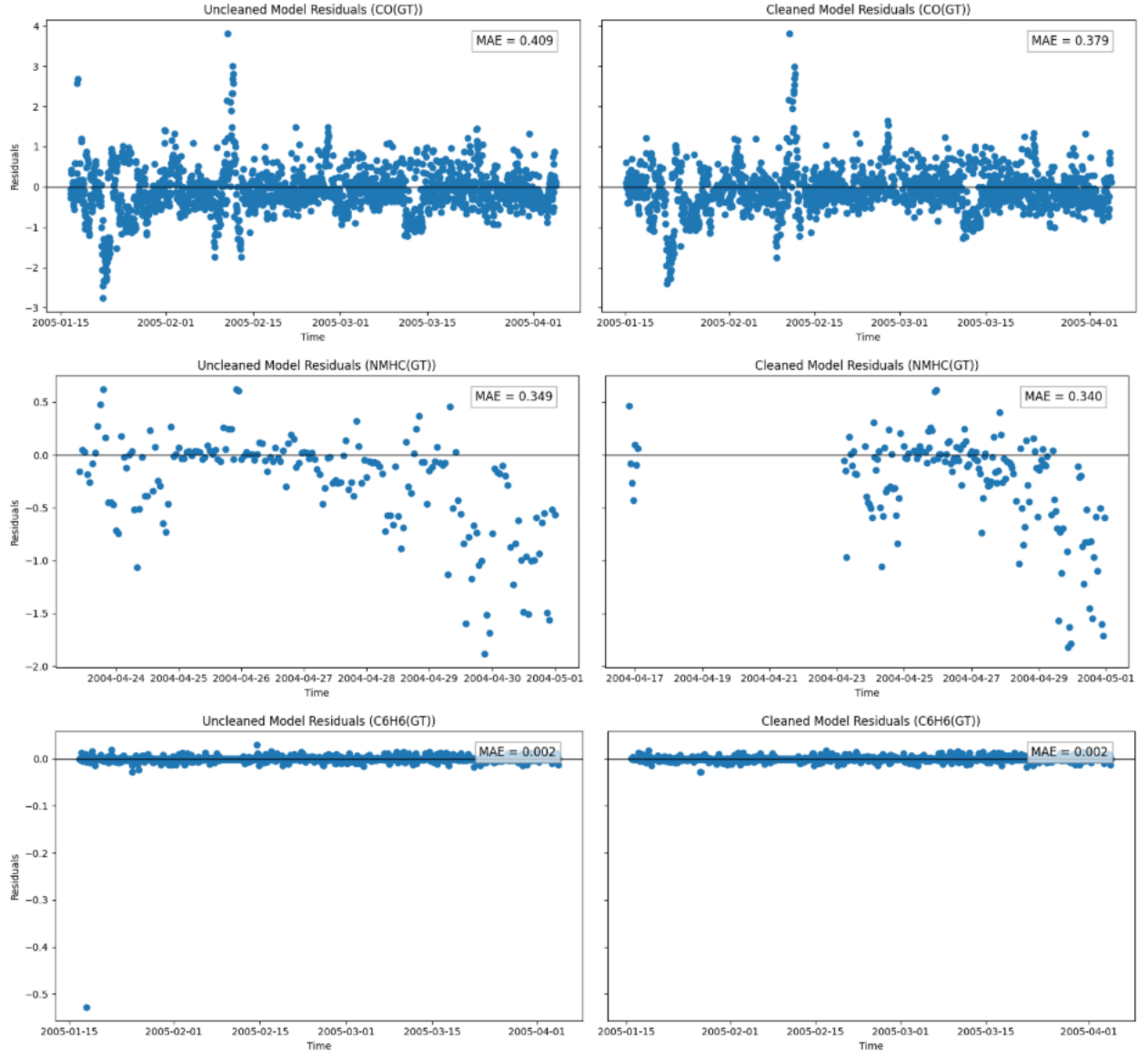


Figure 26: Comparison of two models, one trained on clean dataset with anomalies removed and the other trained on an uncleaned dataset (first 3 targets)

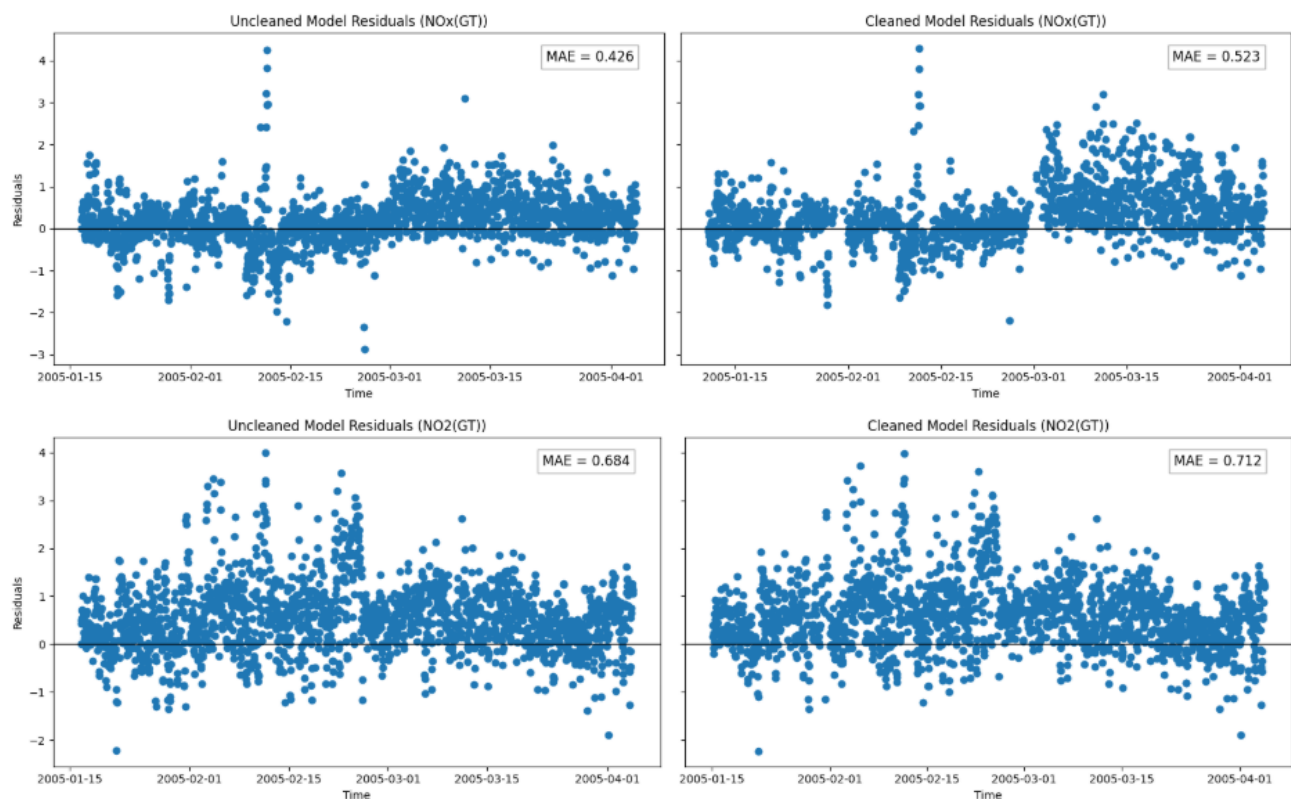


Figure 27: Comparison of two models, one trained on clean dataset with anomalies removed and the other trained on an uncleaned dataset (last 2 targets)

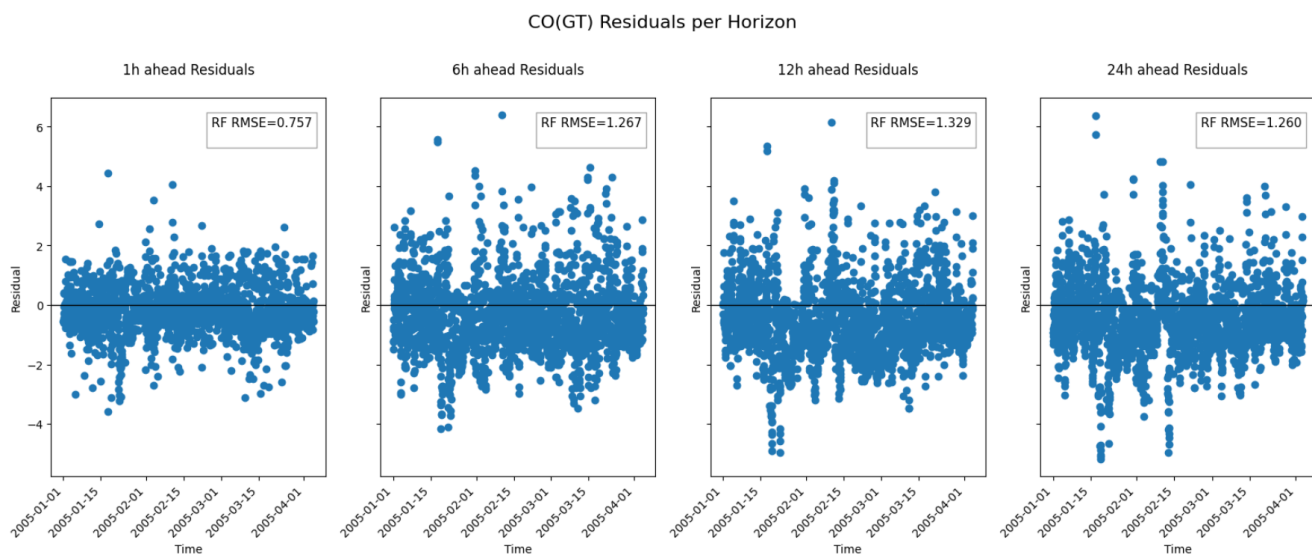


Figure 28: CO(GT) Residuals (Random Forest Regressor)

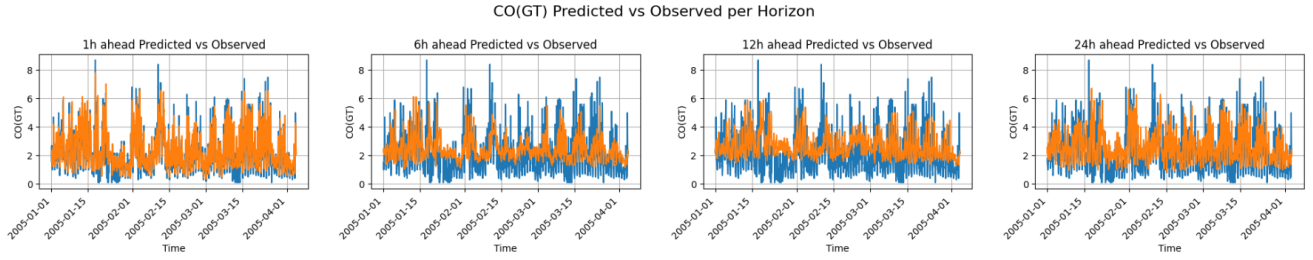


Figure 29: CO(GT) Predictions (orange) vs Observed (blue) (Random Forest Regressor)

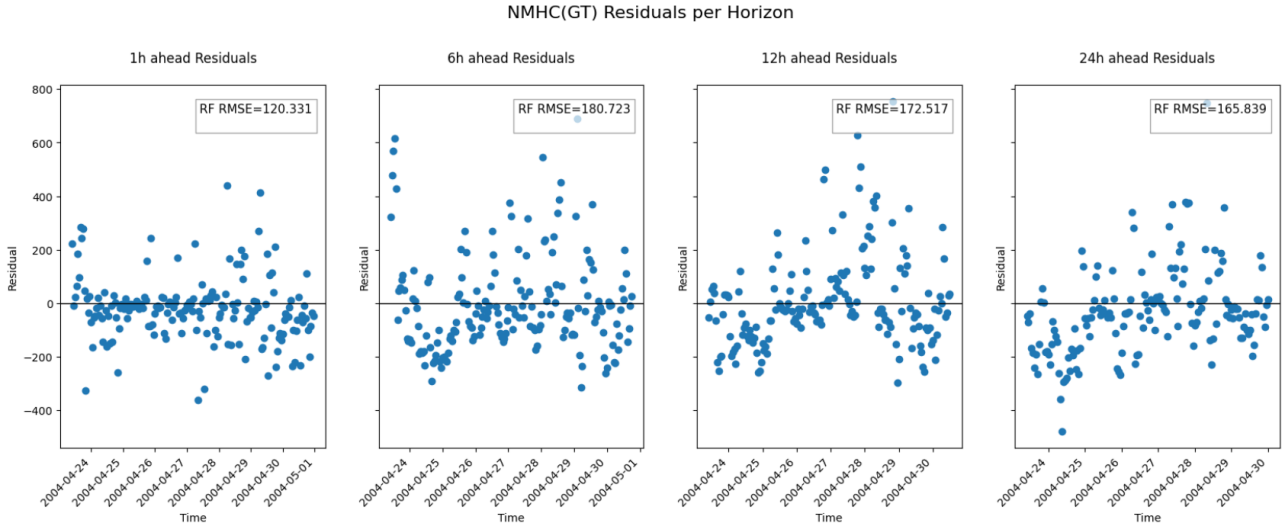


Figure 30: NMHC(GT) Residuals (Random Forest Regressor)

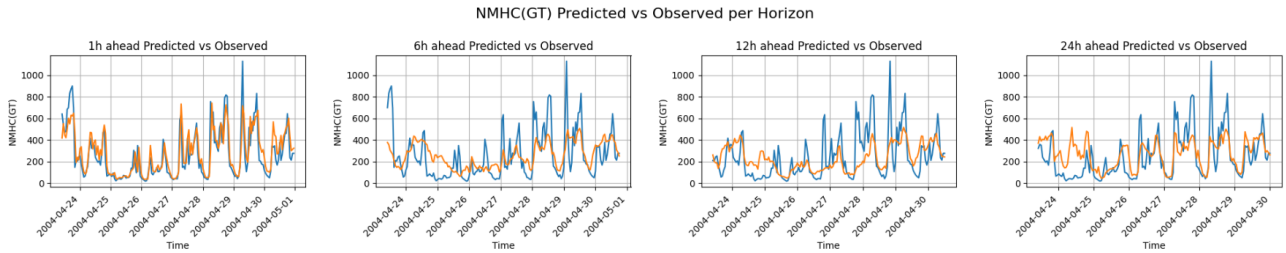


Figure 31: NMHC(GT) Predictions (orange) vs Observed (blue) (Random Forest Regressor)

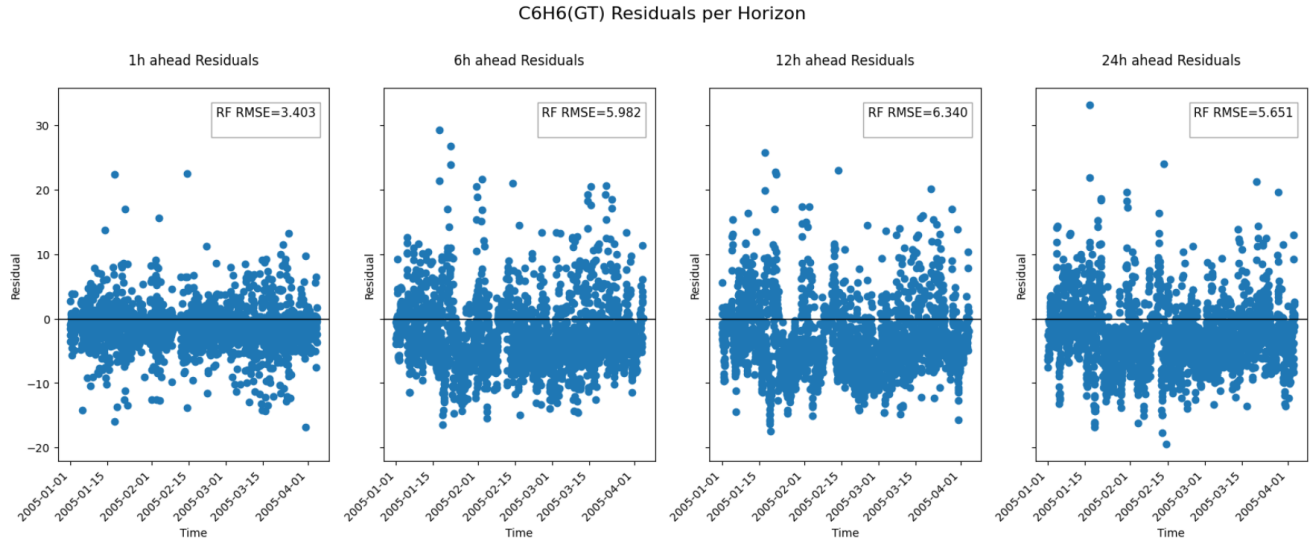


Figure 32: C6H6(GT) Residuals (Random Forest Regressor)

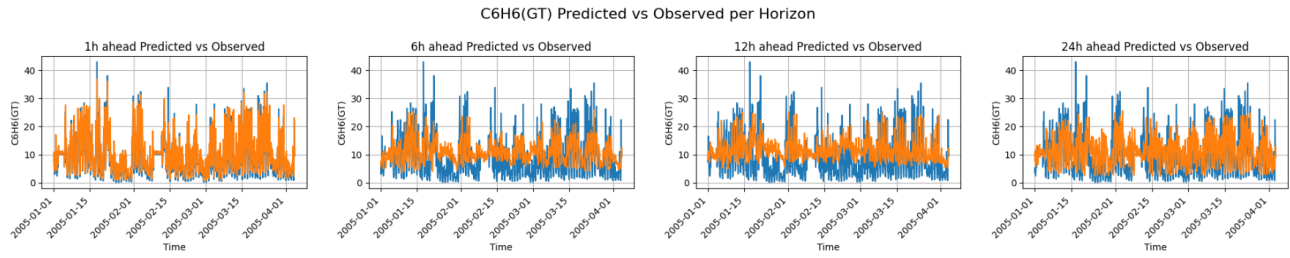


Figure 33: C6H6(GT) Predictions (orange) vs Observed (blue) (Random Forest Regressor)

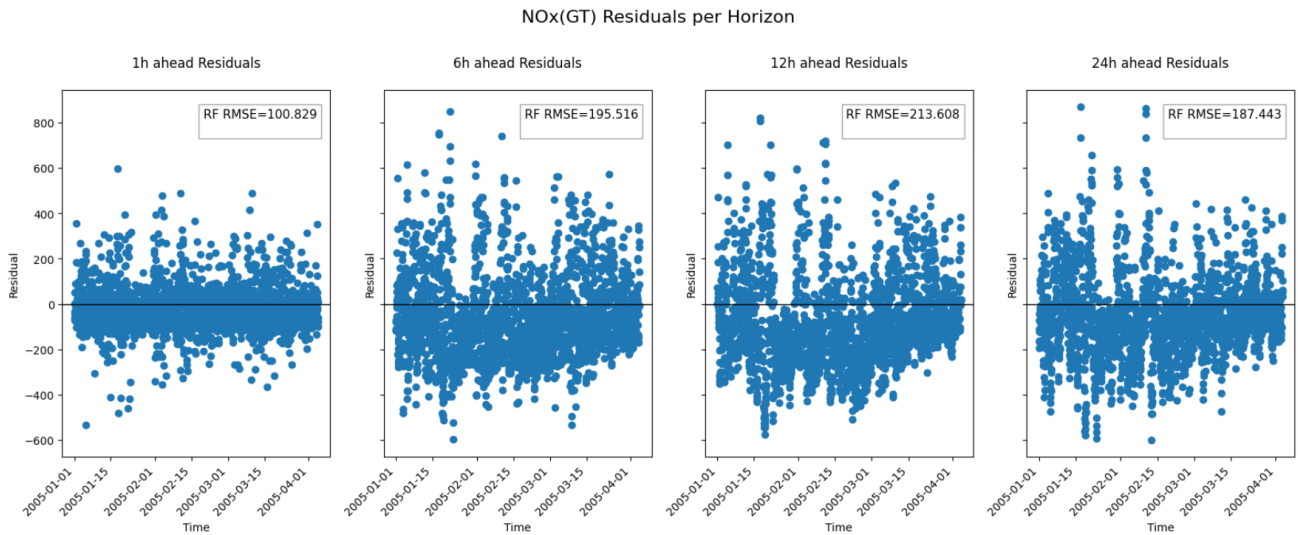


Figure 34: NOx(GT) Residuals (Random Forest Regressor)

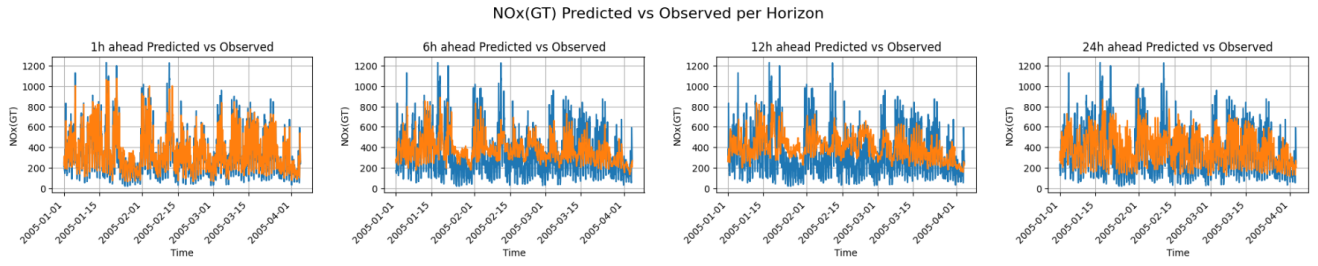


Figure 35: NOx(GT) Predictions (orange) vs Observed (blue) (Random Forest Regressor)

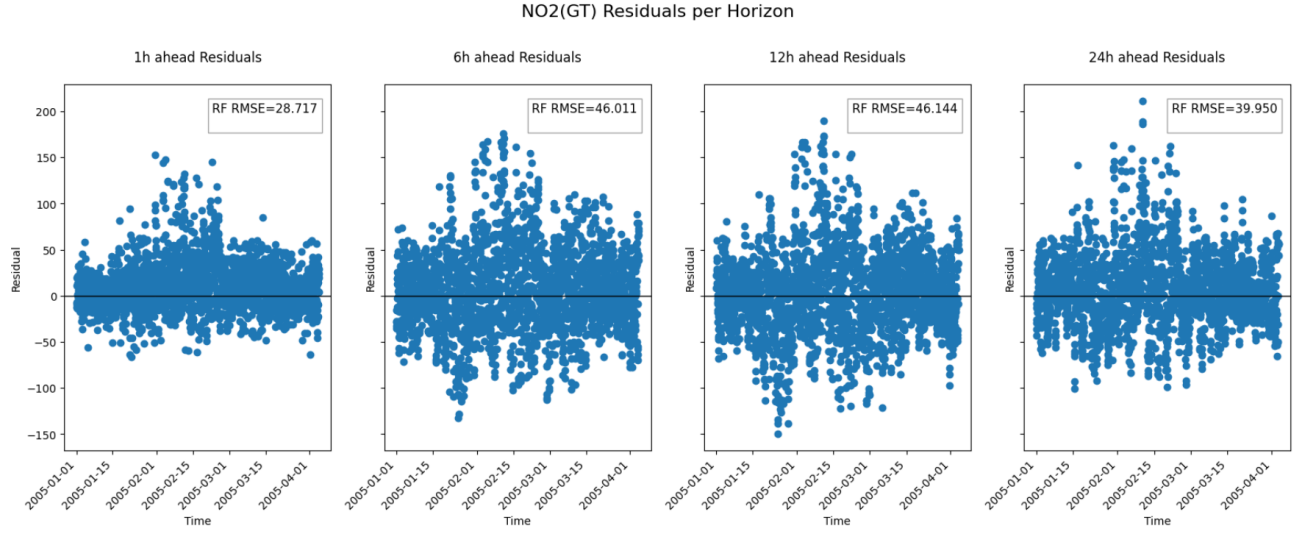


Figure 36: NO2(GT) Residuals (Random Forest Regressor)

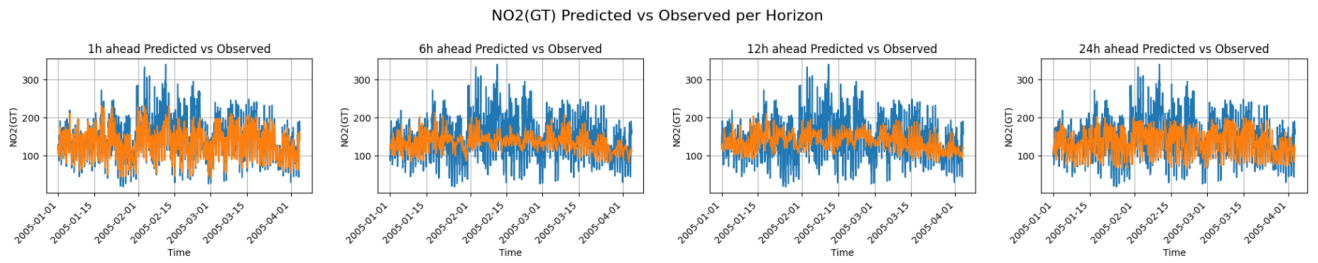


Figure 37: NO2(GT) Predictions (orange) vs Observed (blue) (Random Forest Regressor)

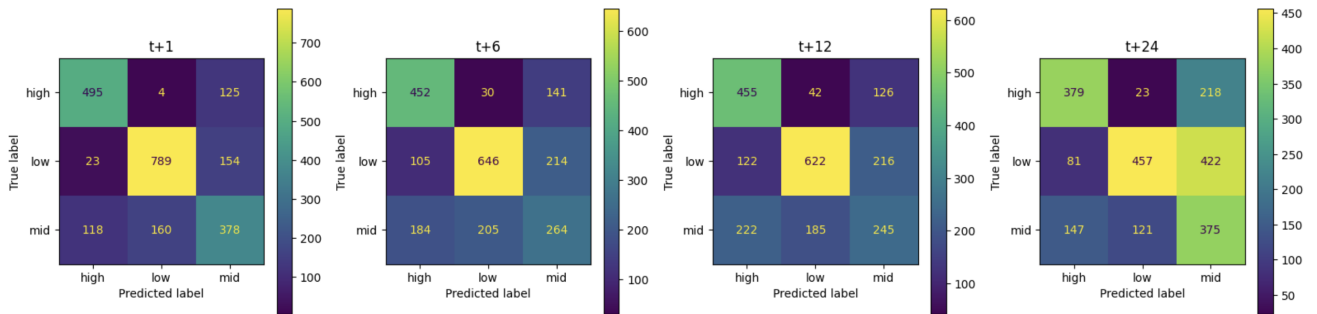


Figure 38: Predicted vs. True Label for CO(GT) Classification on Test Set for Each Horizon

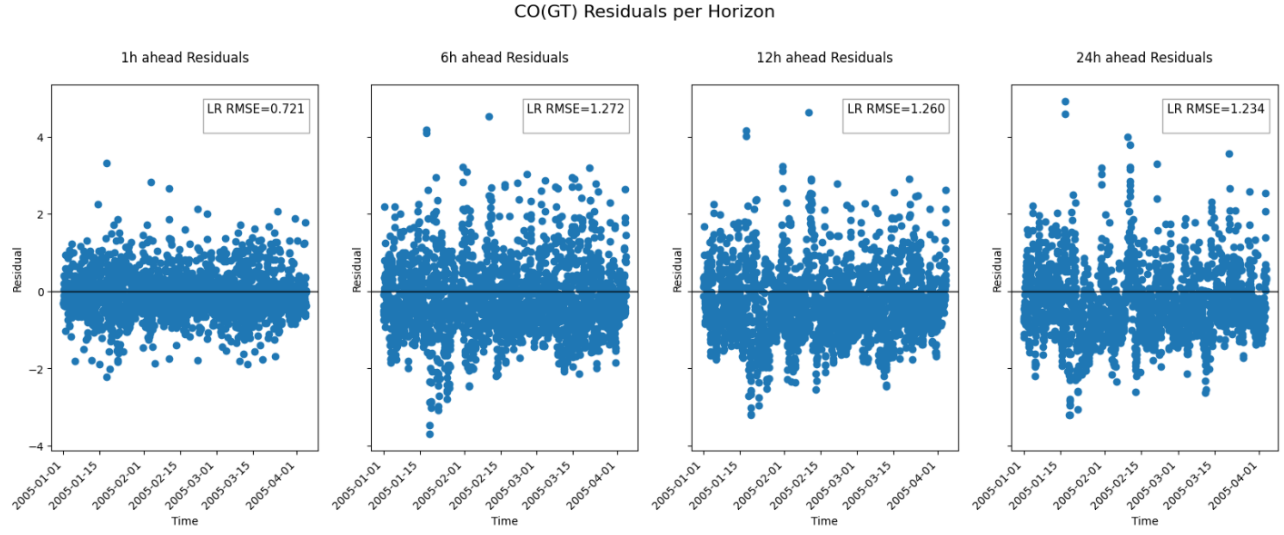


Figure 39: CO(GT) Residuals (Linear Regressor)

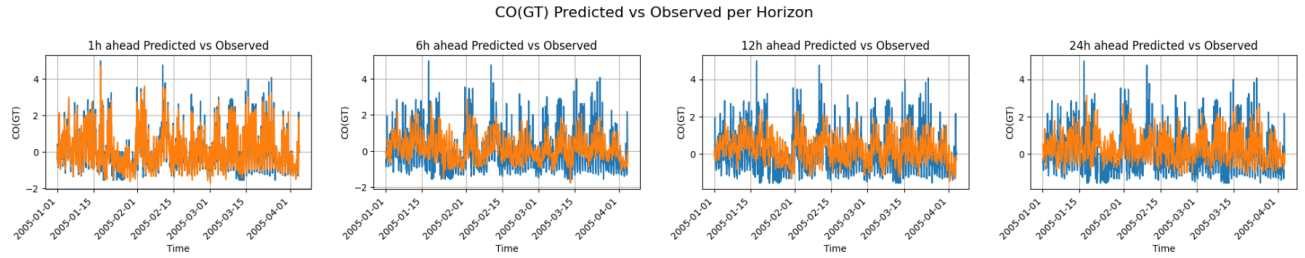


Figure 40: CO(GT) Predictions (orange) vs Observed (blue) (Linear Regressor)

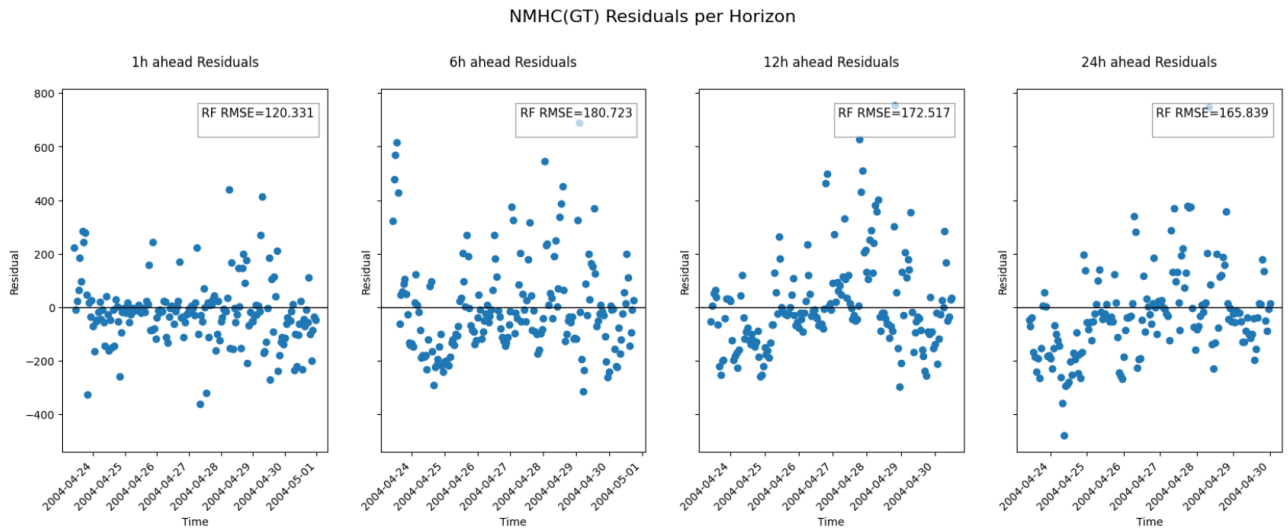


Figure 41: NMHC(GT) Residuals (Linear Regressor)

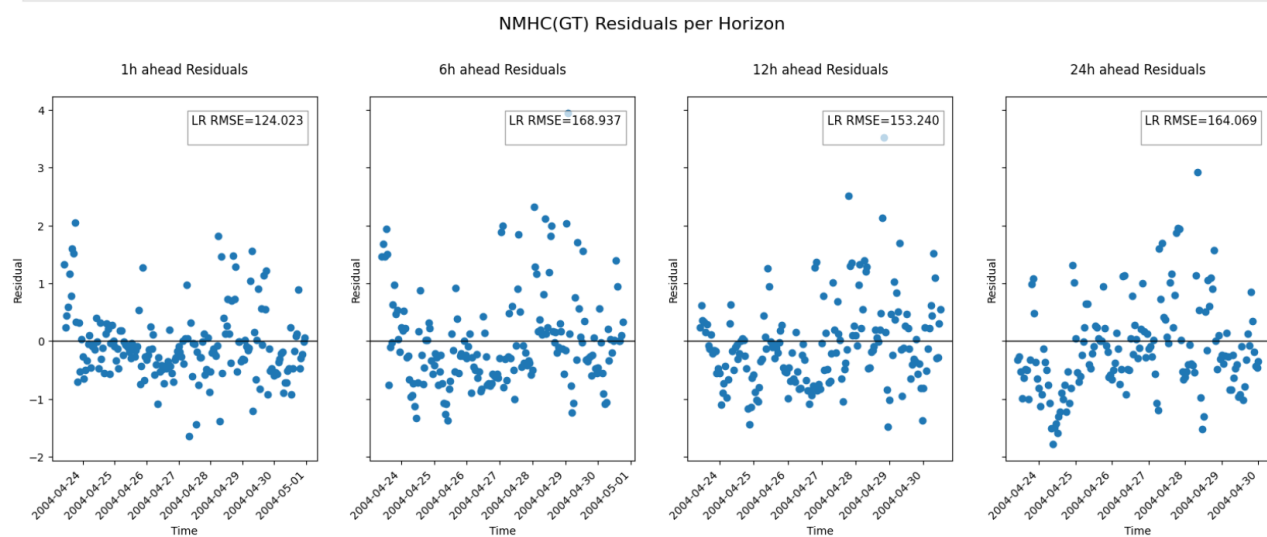


Figure 42: NMHC(GT) Predictions (orange) vs Observed (blue) (Linear Regressor)

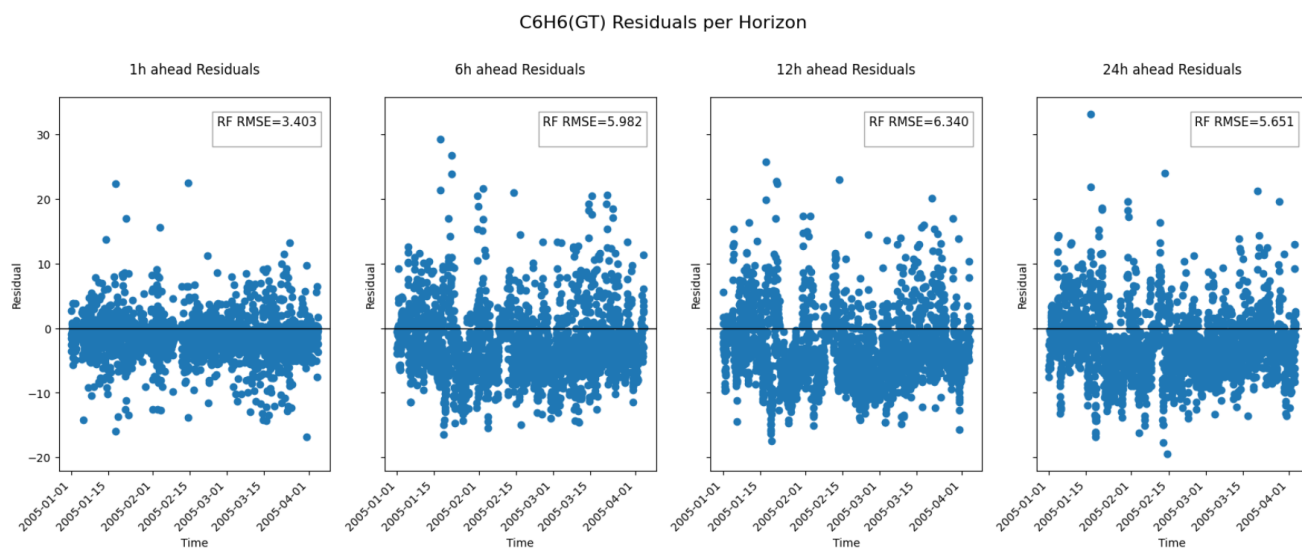


Figure 43: C6H6(GT) Residuals (Linear Regressor)

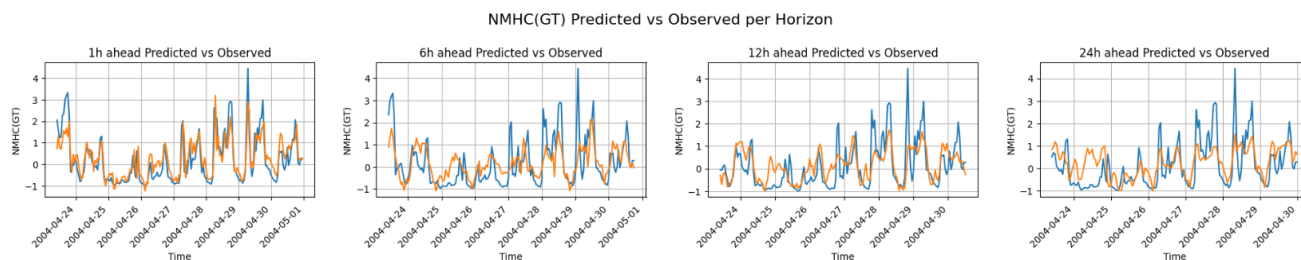


Figure 44: C6H6(GT) Predictions (orange) vs Observed (blue) (Linear Regressor)

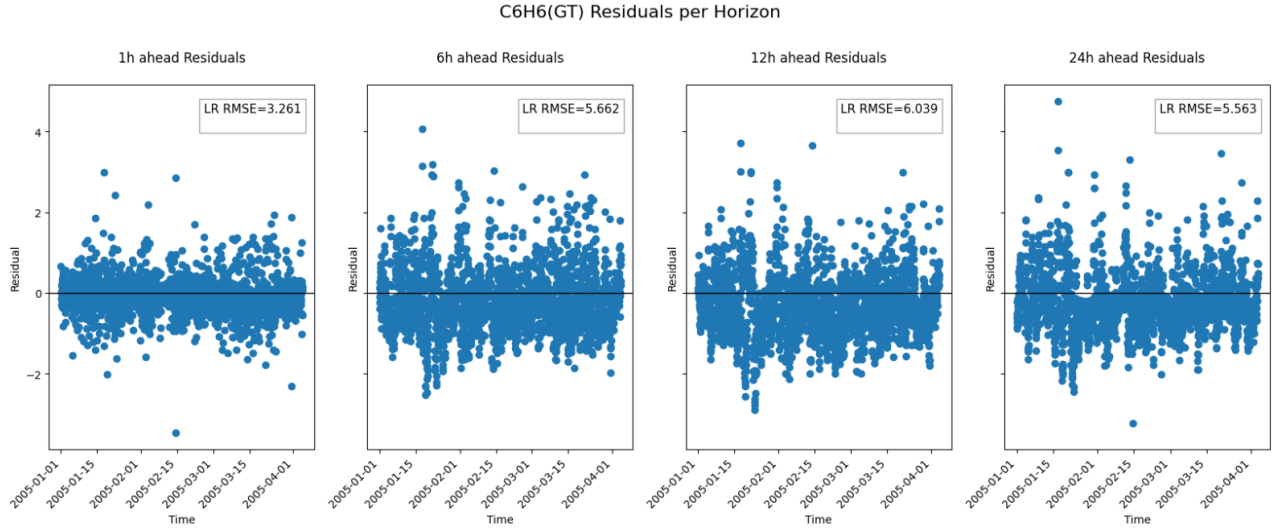


Figure 45: NO_x(GT) Residuals (Linear Regressor)

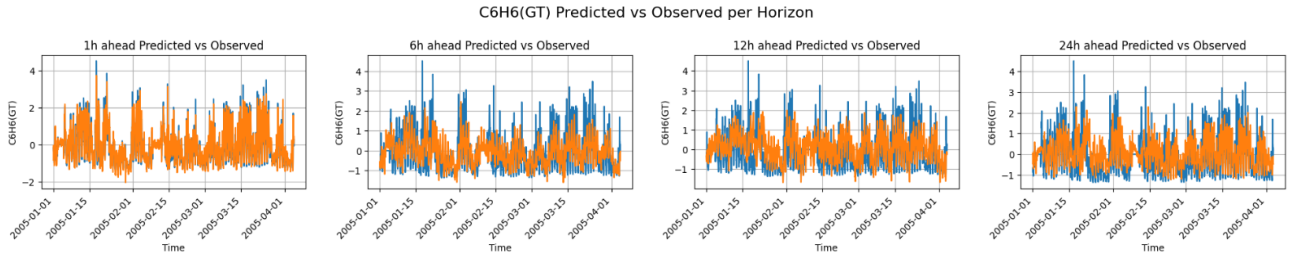


Figure 46: NO_x(GT) Predictions (orange) vs Observed (blue) (Linear Regressor)

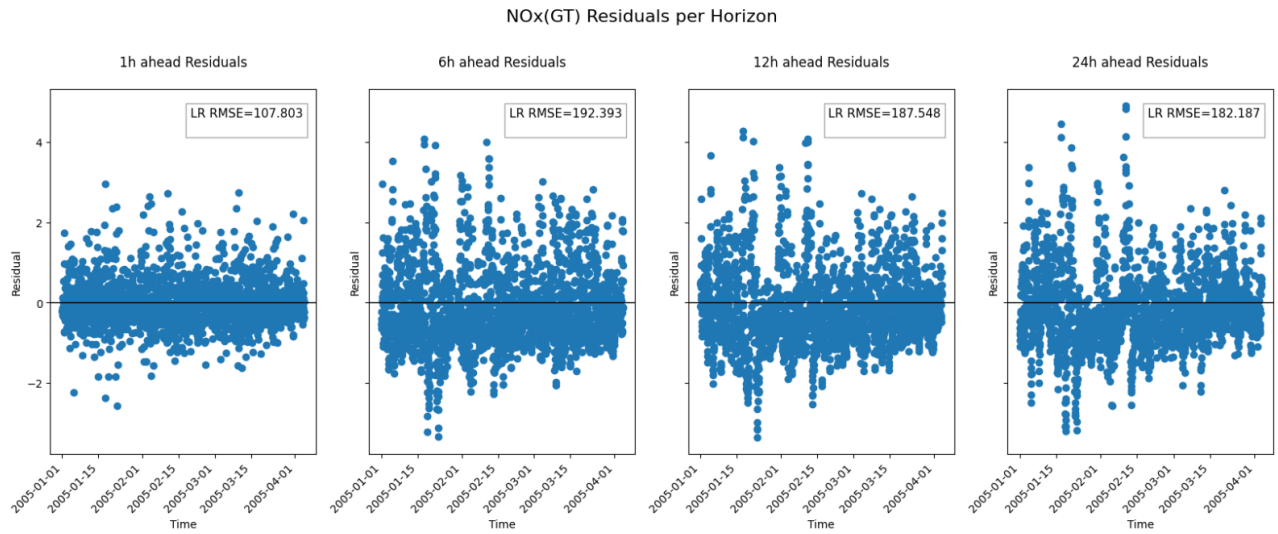


Figure 47: NO_x(GT) Residuals (Linear Regressor)

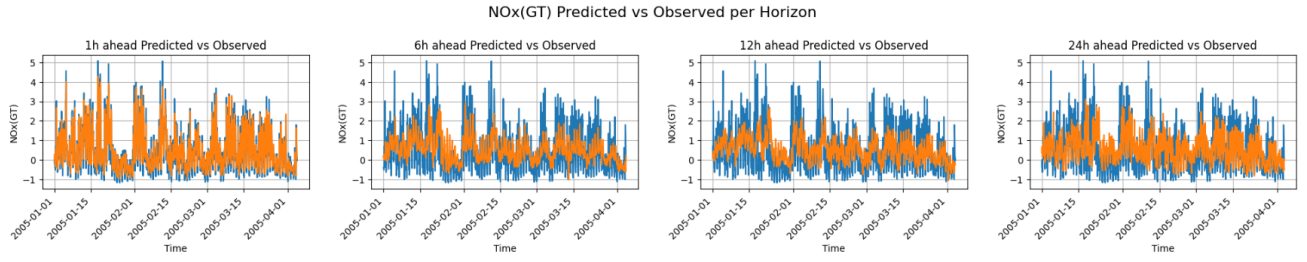


Figure 48: NOx(GT) Predictions (orange) vs Observed (blue) (Linear Regressor)

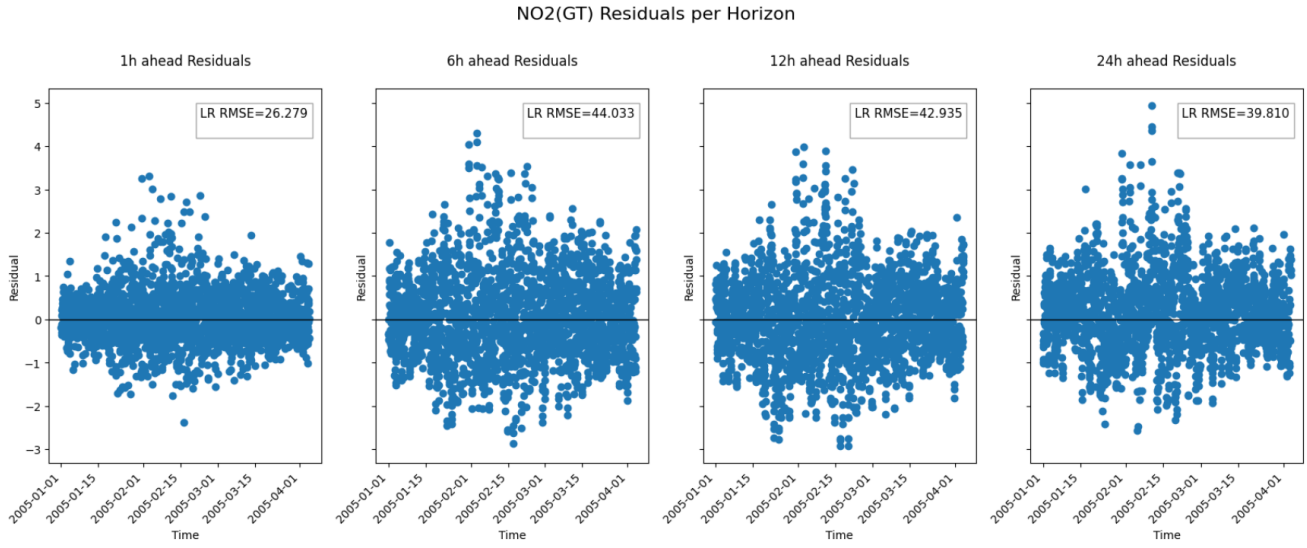


Figure 49: NO2(GT) Residuals (Linear Regressor)

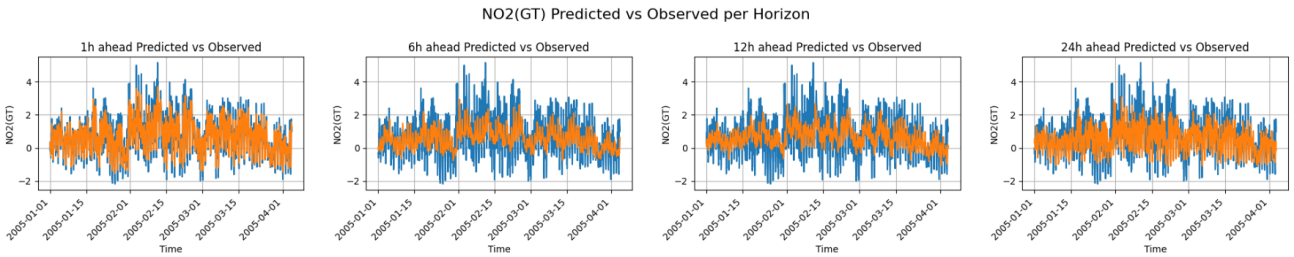


Figure 50: NO2(GT) Predictions (orange) vs Observed (blue) (Linear Regressor)

	high	low	mid
CO(GT)	0.753477	-0.744459	-0.009018
PT08.S1(CO)	0.263517	-0.206372	-0.057145
NMHC(GT)	0.088051	-0.103209	0.015158
C6H6(GT)	0.218305	-0.201228	-0.017076
PT08.S2(NMHC)	0.221165	-0.198422	-0.022743
NOx(GT)	0.371062	-0.471271	0.100208
PT08.S3(NOx)	-0.015049	-0.093506	0.108555
NO2(GT)	0.233197	-0.491371	0.258174
PT08.S4(NO2)	0.181688	-0.221925	0.040237
PT08.S5(O3)	0.106492	-0.015464	-0.091028
T	-0.036225	0.012068	0.024157
RH	-0.024858	0.012119	0.012739
AH	-0.162243	0.057707	0.104536
weekday_0	-0.021428	0.054179	-0.032751
weekday_1	0.044896	-0.107435	0.062539
weekday_2	0.034075	-0.086719	0.052644
weekday_3	0.034375	-0.059079	0.024704
weekday_4	0.124727	-0.043413	-0.081314
weekday_5	-0.073863	0.066281	0.007583
weekday_6	-0.139581	0.174905	-0.035325
hour_first6	-0.374261	0.361674	0.012588
hour_next6	0.122897	-0.214541	0.091644
hour_last6	-0.099084	0.112873	-0.013789
spring	0.022670	-0.090066	0.067396
summer	-0.025356	0.035797	-0.010441
fall	-0.032132	0.048098	-0.015966
winter	0.038019	0.004890	-0.042909
CO(GT)_ma_3	0.013109	-0.080559	0.067450
CO(GT)_ma_6	-0.094570	0.061181	0.033389
CO(GT)_ma_12	0.018301	0.022524	-0.040825
CO(GT)_ma_24	0.286636	-0.196302	-0.090335
CO(GT)_lag1	0.080768	-0.144590	0.063822
NMHC(GT)_lag1	-0.051729	0.044120	0.007609
C6H6(GT)_lag1	-0.247551	0.259481	-0.011931
NOx(GT)_lag1	0.066869	-0.016155	-0.050714
NO2(GT)_lag1	-0.156732	0.083444	0.073288

(a) Coefficients of logistic model for horizon $t + 1$

	high	low	mid
CO(GT)	-0.233468	0.211541	0.021927
PT08.S1(CO)	-0.036382	0.171021	-0.134638
NMHC(GT)	0.057972	-0.105039	0.047067
C6H6(GT)	-0.033068	-1.385003	1.418071
PT08.S2(NMHC)	-0.821662	2.130245	-1.308583
NOx(GT)	0.725851	-0.671260	-0.054590
PT08.S3(NOx)	0.170272	-0.270868	0.100596
NO2(GT)	-0.452377	0.149455	0.302921
PT08.S4(NO2)	0.612264	-0.626154	0.013890
PT08.S5(O3)	0.449151	-0.298952	-0.150199
T	0.398795	-0.421209	0.022414
RH	0.257526	-0.258860	0.001334
AH	-0.689373	0.464157	0.225216
weekday_0	0.325512	-0.248245	-0.077267
weekday_1	0.490182	-0.651886	0.161704
weekday_2	0.169616	-0.311313	0.141697
weekday_3	0.287027	-0.306974	0.019948
weekday_4	0.482996	-0.397460	-0.085536
weekday_5	-0.727257	0.488037	0.239220
weekday_6	-0.894102	0.857663	0.036439
hour_first6	-0.092560	0.224763	-0.132203
hour_next6	-0.782006	0.614273	0.167733
hour_last6	-1.915737	1.585094	0.330643
spring	0.035517	-0.344286	0.308769
summer	-0.215227	0.148617	0.066610
fall	0.149930	-0.187379	0.037449
winter	0.163754	-0.187130	0.023376
CO(GT)_ma_3	0.461445	-0.500809	0.039364
CO(GT)_ma_6	-0.262044	0.558881	-0.296837
CO(GT)_ma_12	-1.182720	1.154965	0.027755
CO(GT)_ma_24	2.108921	-2.078997	-0.029924
CO(GT)_lag1	-0.280787	0.243954	0.036834
NMHC(GT)_lag1	-0.031195	0.037191	-0.005996
C6H6(GT)_lag1	0.083391	0.133000	-0.216391
NOx(GT)_lag1	-0.097690	-0.058116	0.155806
NO2(GT)_lag1	0.171159	-0.286751	0.115592

(b) Coefficients of logistic model for horizon $t + 6$

Figure 51: Coefficient plots of the logistic regression model for horizons $t + 1$ and $t + 6$.

	high	low	mid
CO(GT)	-0.255608	0.233706	0.021902
PT08.S1(CO)	-0.075403	0.159675	-0.084272
NMHC(GT)	-0.020546	0.000062	0.020484
C6H6(GT)	-0.225312	0.578275	-0.352963
PT08.S2(NMHC)	-0.158097	0.118397	0.039700
NOx(GT)	0.183610	-0.127675	-0.055935
PT08.S3(NOx)	-0.135719	-0.149614	0.285333
NO2(GT)	-0.302055	0.111691	0.190364
PT08.S4(NO2)	0.584211	-0.514931	-0.069280
PT08.S5(O3)	0.699080	-0.569672	-0.129408
T	-0.423853	0.156977	0.266876
RH	0.014217	-0.058723	0.044505
AH	-0.267640	0.190727	0.076913
weekday_0	0.301400	-0.133425	-0.167975
weekday_1	0.267526	-0.414238	0.146712
weekday_2	0.135851	-0.222247	0.086396
weekday_3	0.354964	-0.320959	-0.034005
weekday_4	0.083850	-0.023401	-0.060449
weekday_5	-0.875064	0.727583	0.147481
weekday_6	-0.300152	0.402452	-0.102300
hour_first6	0.653761	-0.641230	-0.012531
hour_next6	1.121448	-1.107260	-0.014188
hour_last6	1.044908	-0.777202	-0.267706
spring	-0.350541	0.106524	0.244017
summer	-0.145776	0.248438	-0.102662
fall	0.332961	-0.259327	-0.073634
winter	0.131731	-0.079871	-0.051860
CO(GT)_ma_3	0.284255	-0.319601	0.035346
CO(GT)_ma_6	-0.939230	0.903809	0.035421
CO(GT)_ma_12	1.110073	-1.048099	-0.061975
CO(GT)_ma_24	0.422758	-0.271784	-0.150974
CO(GT)_lag1	0.229466	-0.111104	-0.118362
NMHC(GT)_lag1	-0.013686	0.017025	-0.003339
C6H6(GT)_lag1	-0.267395	0.223864	0.043531
NOx(GT)_lag1	-0.237361	0.190222	0.047139
NO2(GT)_lag1	0.017135	-0.195052	0.177917

(c) Coefficients of logistic model for horizon $t + 12$

	high	low	mid
CO(GT)	0.361949	-0.505136	0.143186
PT08.S1(CO)	0.232942	-0.184179	-0.048762
NMHC(GT)	0.008250	-0.066928	0.058679
C6H6(GT)	0.114741	-0.095682	-0.019060
PT08.S2(NMHC)	0.131931	-0.103404	-0.028527
NOx(GT)	0.062664	0.033220	-0.095884
PT08.S3(NOx)	-0.132630	0.029248	0.103381
NO2(GT)	0.069210	-0.067323	-0.001887
PT08.S4(NO2)	0.016501	0.008155	-0.024656
PT08.S5(O3)	0.040986	0.111346	-0.152332
T	-0.220709	0.107894	0.112816
RH	-0.165751	0.055019	0.110732
AH	-0.110394	0.035905	0.074489
weekday_0	0.204057	-0.192703	-0.011354
weekday_1	0.023971	-0.150914	0.126943
weekday_2	0.142540	-0.163663	0.021123
weekday_3	0.216454	-0.173953	-0.042501
weekday_4	-0.395708	0.348915	0.046794
weekday_5	-0.454793	0.489094	-0.034301
weekday_6	0.264086	-0.161483	-0.102603
hour_first6	-0.422362	0.433126	-0.010764
hour_next6	0.050259	-0.016022	-0.034238
hour_last6	0.159480	-0.096629	-0.062850
spring	-0.069360	-0.023203	0.092563
summer	-0.113207	0.149509	-0.036303
fall	0.100519	-0.084474	-0.016045
winter	0.082655	-0.046540	-0.036115
CO(GT)_ma_3	0.013663	-0.018145	0.004483
CO(GT)_ma_6	-0.025938	-0.070856	0.096795
CO(GT)_ma_12	0.011969	0.039313	-0.051282
CO(GT)_ma_24	-0.047638	0.208313	-0.160675
CO(GT)_lag1	0.117948	-0.044769	-0.073179
NMHC(GT)_lag1	0.023039	-0.048302	0.025263
C6H6(GT)_lag1	0.087118	-0.066052	-0.021066
NOx(GT)_lag1	0.065274	-0.072478	0.007204
NO2(GT)_lag1	0.007537	-0.239968	0.232431

(d) Coefficients of logistic model for horizon $t + 24$

Figure 52: Coefficient plots of the logistic regression model for horizons $t + 12$ and $t + 24$.